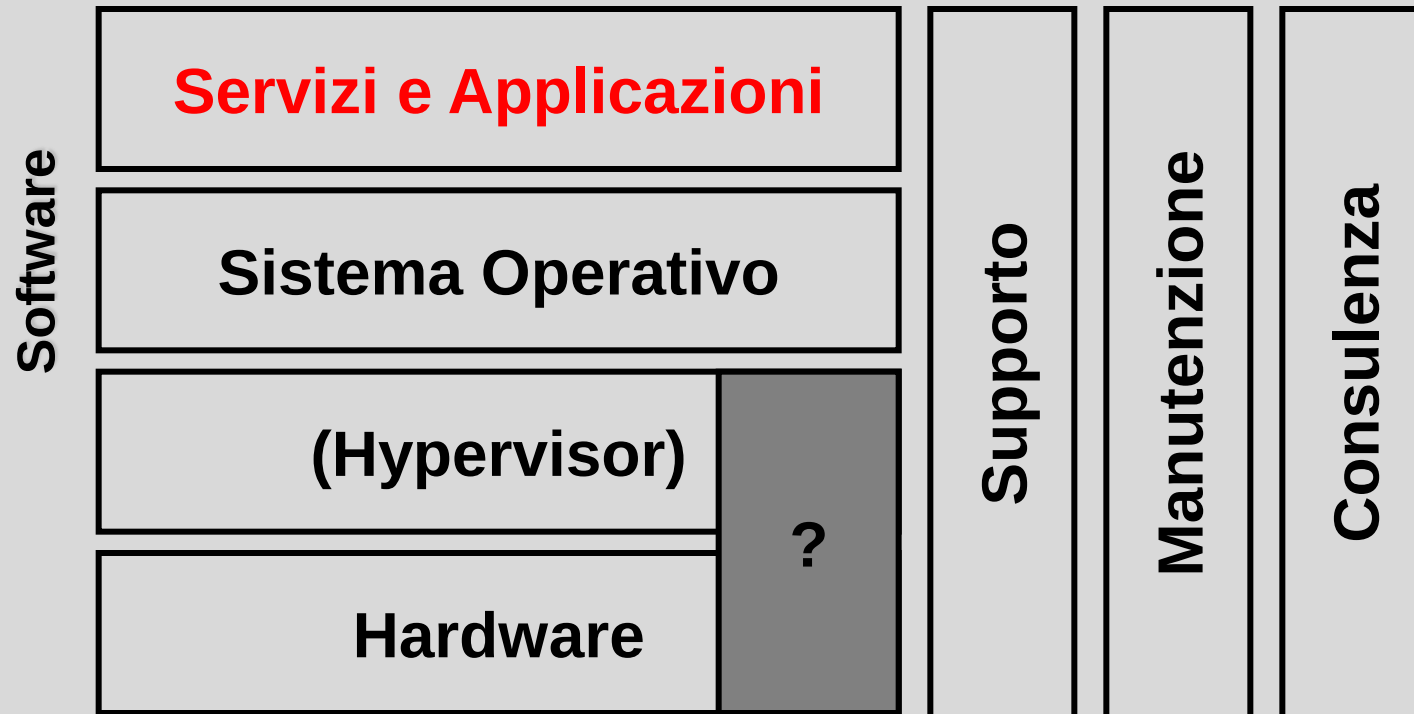


Amministrazione di Sistema

Server

Il “sistema server”



Chi gestisce cosa?

Supporto e certificazioni

- Supporto e garanzia
 - Sull'hardware
 - NDB, 4h, Prosupport, ...
 - Sul software
 - ...
- Certificazioni / Supportabilità
 - Hardware e sistema operativo
 - Hardware e hypervisor
 - Hypervisor e sistema operativo
 - Applicazioni e sistema operativo

Contratti di manutenzione

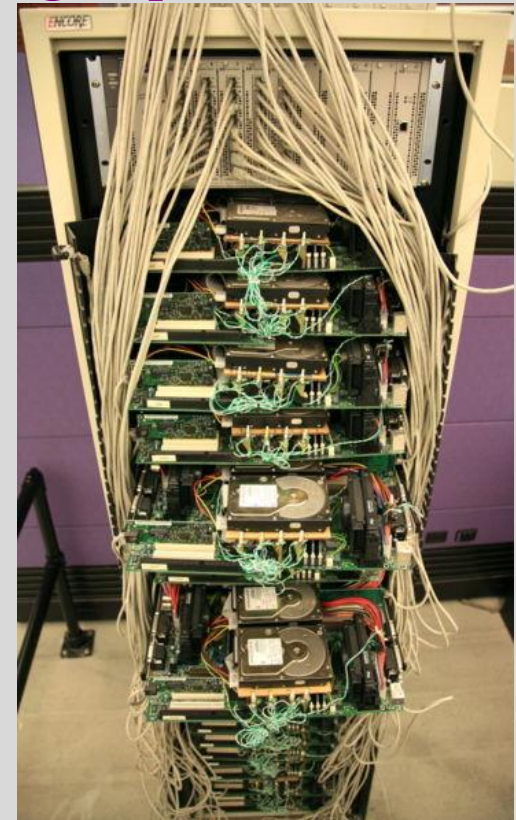
- Caratteristiche
 - On-site
 - Durata
 - Tempistiche
- Hardware
- Software e/o servizi
- Importanza della manutenzione preventiva

Il “sistema server”

- Soluzioni “standard”
 - COTS (Commercial Off-The-Shelf) component
- Soluzioni “custom”
 - http://en.wikipedia.org/wiki/Google_platform



- Soluzioni “cloud”

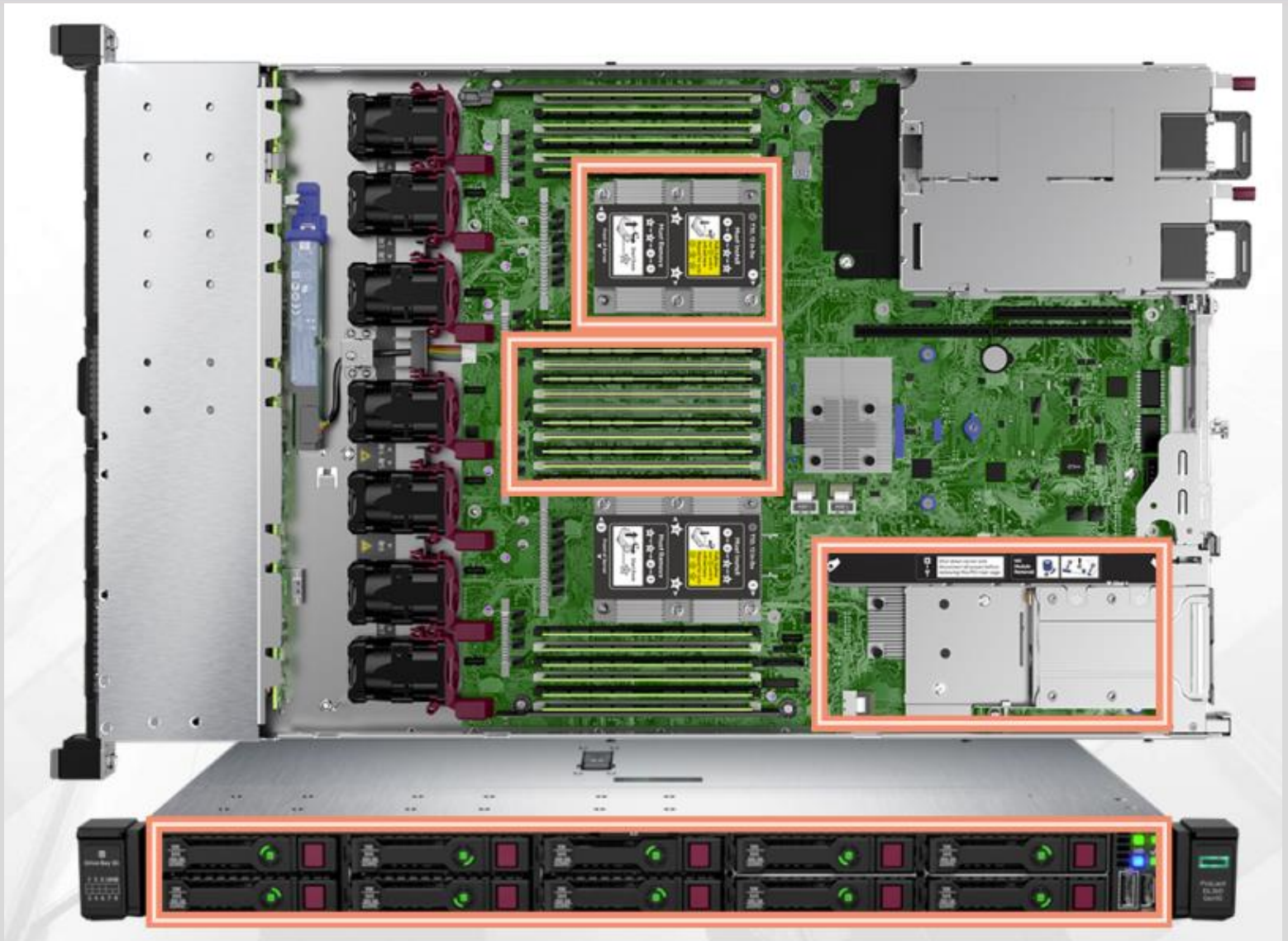


Form factor

- Formato rack
- Formato tower
- Formato «modulare»
- Altri formati
 - Embedded/Mini
 - Industriale
 - Rugged
 - IoT

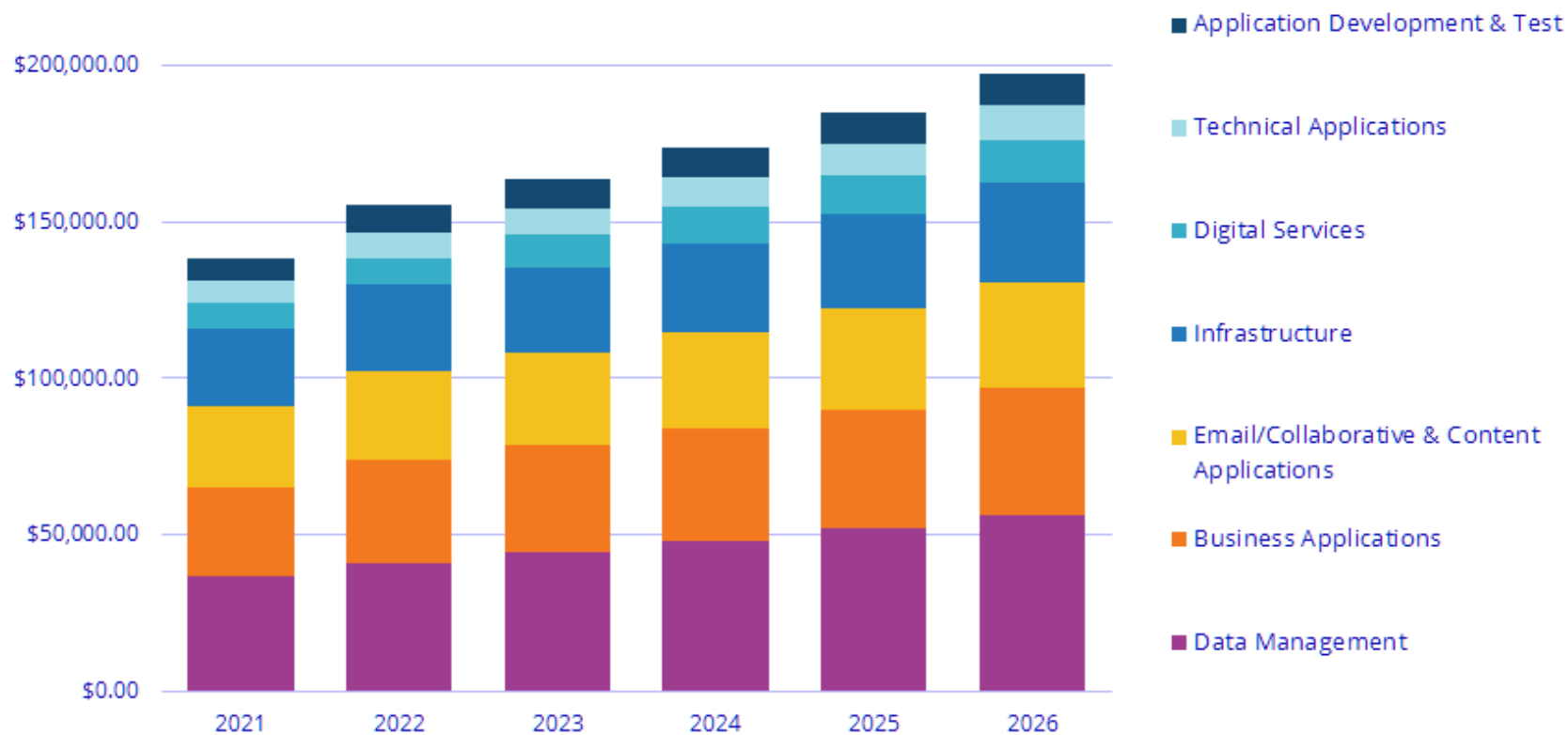


Server rack



Crescita del mondo server

Worldwide Enterprise Infrastructure Workloads Forecast, 2021 - 2026 (Value US\$M)



Source: IDC 2022

Server vendors

Top 5 Companies, Worldwide Server Vendor Revenue, Market Share, and Growth, Third Quarter of 2020 <i>(Revenues are in US\$ Millions)</i>					
Company	3Q20 Revenue	3Q20 Market Share	3Q19 Revenue	3Q19 Market Share	3Q20/3Q19 Revenue Growth
T1. Dell Technologies*	\$3,757.8	16.65%	\$3,779.1	17.12%	-0.6%
T1. HPE/New H3C Group**	\$3,596.9	15.94%	\$3,737.9	16.93%	-3.8%
3. Inspur/Inspur Power Systems ^b	\$2,114.7	9.37%	\$1,973.3	8.94%	7.2%
4. Lenovo	\$1,326.0	5.88%	\$1,189.8	5.39%	11.4%
5. Huawei	\$1,098.9	4.87%	\$916.7	4.15%	19.9%
ODM Direct	\$6,303.3	28.03%	\$5,816.0	26.34%	8.4%
Rest of Market	\$4,367.9	19.36%	\$4,663.8	21.13%	-6.3%
Total	\$22,565.6	100.00%	\$22,076.6	100.00%	2.2%
Source: IDC Worldwide Quarterly Server Tracker, December 8, 2020					

Tipi di server

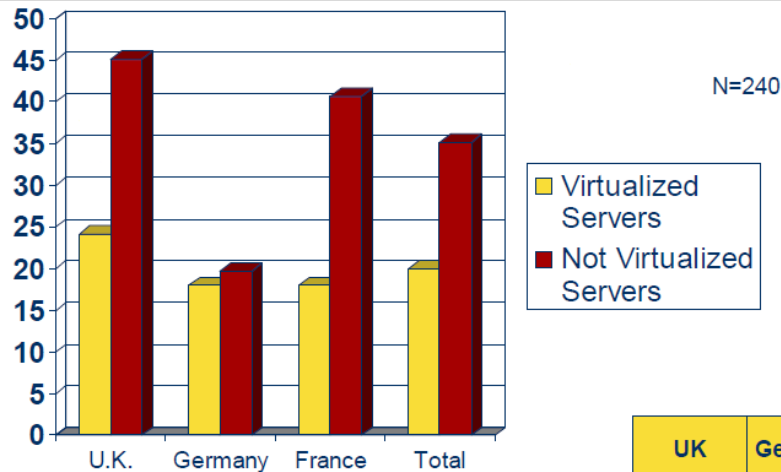
- In base all'architettura
- In base al Sistema Operativo
- In base ai servizi offerti
 - File Server
 - Mail Server
 - DNS Server
 - Web Server
 - ...
- In base alle risorse usate/richieste
 - CPU intensive
 - RAM intensive
 - Storage o I/O intensive

I “server”

- Tipicamente più di uno
 - Necessità di sistemi operativi diversi
 - Necessità di servizi diversi
 - Necessità di disponibilità elevata
 - Necessità di prestazioni adeguate
 - ...
- Scale-in vs Scale-out
 - Scalabilità verticale oppure orizzontale?
 - Meglio pochi server grossi o tanti piccoli?
 - Risorse «aggregate» o «disaggregate»?
 - Off-load di alcune operazioni?

Server fisici vs Server virtuali

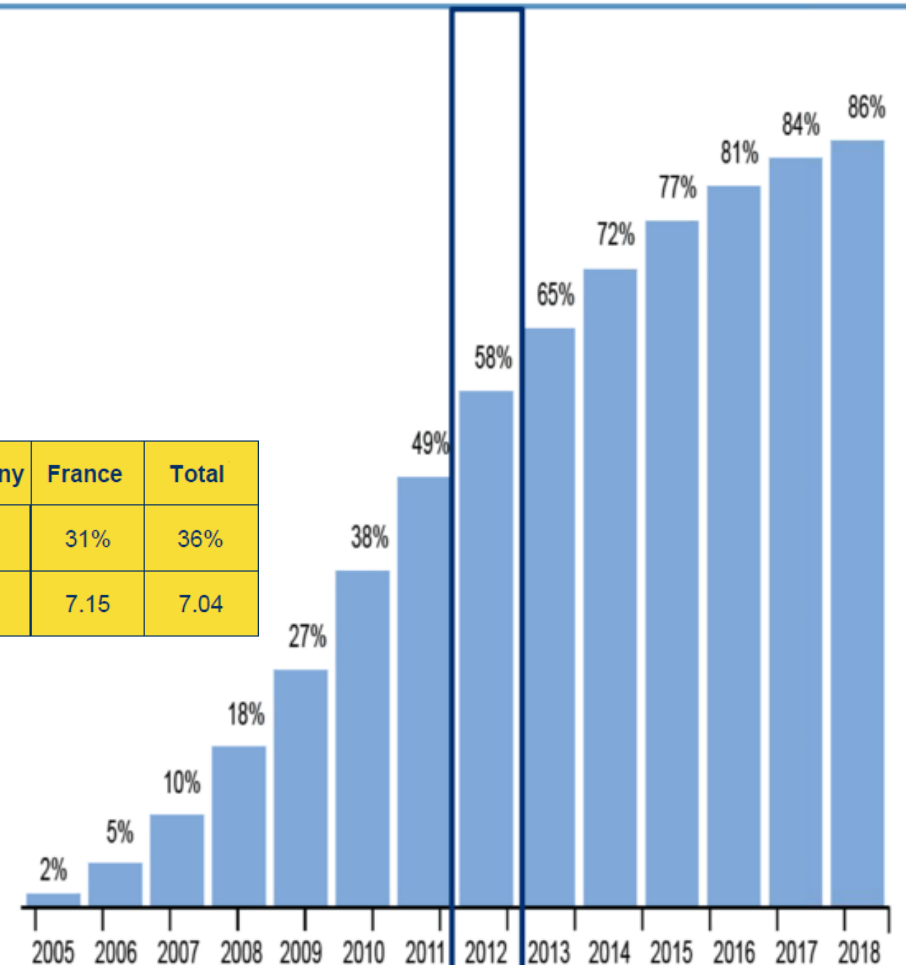
Figure 1. Percentage of x86-Architecture Workloads Running in VMs



	UK	Germany	France	Total
Percentage of virtualized servers	35%	48%	31%	36%
Mean average no. of partition /servers	7.35	6.63	7.15	7.04

IDC 2009

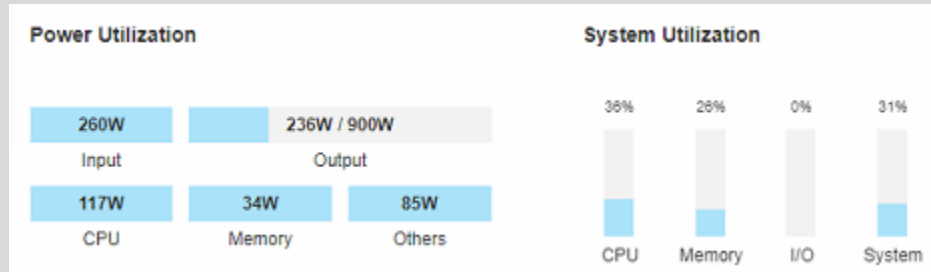
- Nel 2009 ci doveva essere il sorpasso
- In ogni caso la virtualizzazione server è diventata mainstream



Source: Gartner (March 2011)

Efficienza energetica e termica

- Consumo di un server moderno



- Nuovi trend

- Fresh air cooling
 - Server, storage e apparati di rete costruiti per sopportare escursioni termiche a breve termine fino a 45°C
- Performance per Watt
- Data center infrastructure efficiency (DCIE)
- SNIA Emerald
 - ISO standard ISO/IEC 24091:2019 Information technology – Power efficiency measurement specification for data center storage

Tipi di Sistemi Operativi

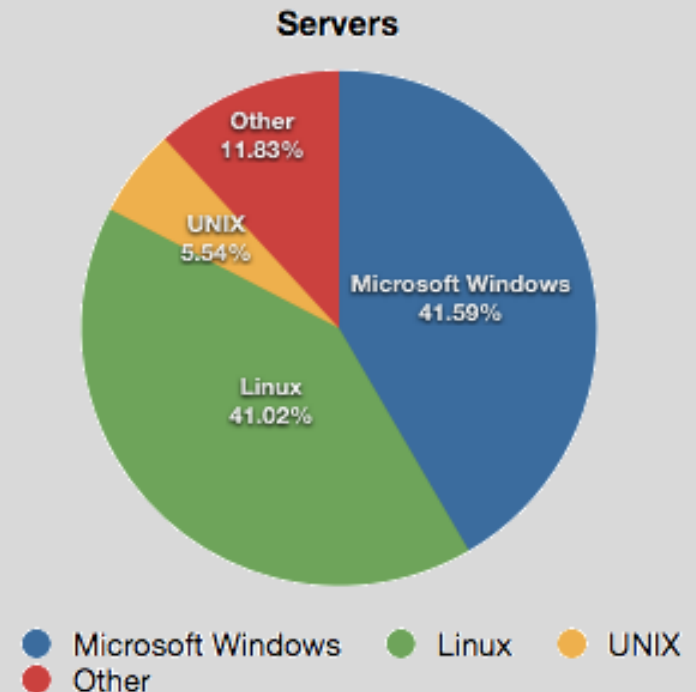
- Oramai tutti i SO sono anche SO di rete (NOS)
- In base alle funzionalità di rete
 - Client/Workstation
 - Server
- In base al tipo di licenza
 - Software proprietario
 - Software libero
- ...



Sistemi Operativi Server

– Server

- Windows Server
 - (NT, 2000)
 - (2003, 2008, 2008R2, 2012, 2012R2)
 - 2016, 2019, 2022
- Linux
 - Distribuzioni varie
- Unix
 - Mac OS X
(futuro della piattaforma server?)
 - FreeBSD
 - NetBSD
 - Solaris
- Altro
 - (Novell Netware)
 - Sistemi operativi di mini/mainframe



Linux vs Windows

- Linux è più stabile?
- Linux o Windows è più veloce?
- Windows è lo standard di mercato?
- Windows è più semplice?
 - La curva di apprendimento è ancora vera?
- Un sistema è più sicuro di un altro?
 - Confronto tra il numero di patch?
 - Quali patch?
- Obsolescenza hardware in relazione al sistema
 - Hardware nuovo può richiede OS recenti
 - Hardware obsoleto può richiede OS vecchi
- Cosa costa di più?
 - confronto TCO (Total Cost of Ownership)

Caratteristiche di un server

- Affidabilità (Reliability), stabilità (Stability), disponibilità (Availability) e ripristinabilità (Recoverability)
 - un server (e i suoi servizi) devono funzionare!
 - MTBC (Mean Time Between Crashes)
 - MTBF (Mean Time Between Failures)
- Scalabilità (Scalability)
 - adattare il server in base al “carico di lavoro”
- Prestazioni (Performance)
 - i tempi di attesa non devono essere eccessivi
- Manutenibilità (Manageability) e semplicità
 - deve essere semplice ma anche efficace configurare e gestire il sistema
- Sicurezza (Security)
- Altro ...
 - Disponibilità di driver
 - Tipo di licenza e tipo di assistenza

Software sui server

- Sistema Operativo (Server)
- Applicazioni tradizioni
 - Programmi di backup
 - Programmi di gestione
- Applicazioni client-server
 - Servizi (demoni)
 - Lato “server” di applicazioni note
- Applicazioni web-oriented

Chi lavora sui server?

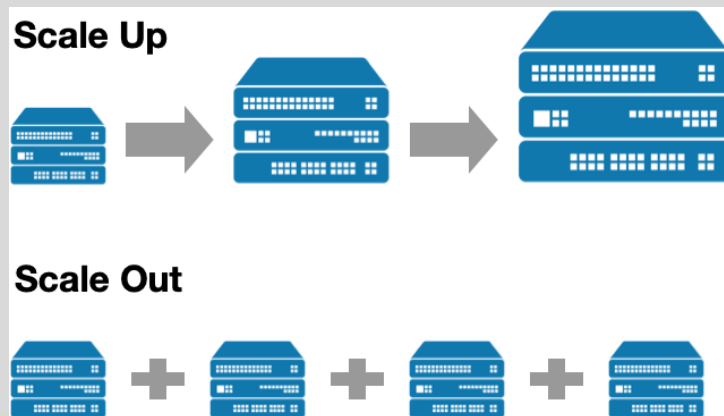
- Amministratore di sistema
 - Persona (o più) fisica
 - Organizzazione interna e/o Azienda esterna
 - Rapporto tipico tra server e sistemisti di 20-30:1
 - per ogni dollaro speso per server e altro hardware, 8 dollari sono destinati a coprire i costi di gestione correlati
 - nel caso di server fisici
- Tecnici
- Programmatori
- Utenti
- Analisti
- A che livello opera ciascuna figura?

La scelta dei server (1)

- Selezione hardware e software
 - Diversi criteri di scelta
 - devono soddisfare i requisiti di un sistema server
 - Privilegiare soprattutto la stabilità
 - hardware e software consolidati
- Scelta hardware
 - CPU, RAM, Storage, Rete, ...
- Scelta software
 - Sistema Operativo, Applicazioni, Servizi, ...

La scelta dei server (2)

- Dimensionamento del sistema o dell'impianto
 - Soluzione verticale (scale-in)
 - server molto potente (limiti di risorse massime)
 - Soluzione orizzontale (scale-out)
 - tanti “piccoli” server
 - sistemi distribuiti
 - Web-scale / Cloud-scale / Hyper-scale
 - Soluzione ottimale?



Sistemi Composabile

Sistemi Distribuiti

La scelta dei server (3)

- Soluzioni per incrementare l'affidabilità e/o le prestazioni
 - Multi-processore (SMP)
 - Controller RAID o storage Enterprise
 - Alimentatori ridondati
 - Sistemi “cluster”
 - cluster “fisico”
 - macchine in ridondanza tra loro
 - Sistema operativo distribuito o per cluster
 - cluster “logico”
 - servizi ridondati tramite DNS o NAT

Scelta del software

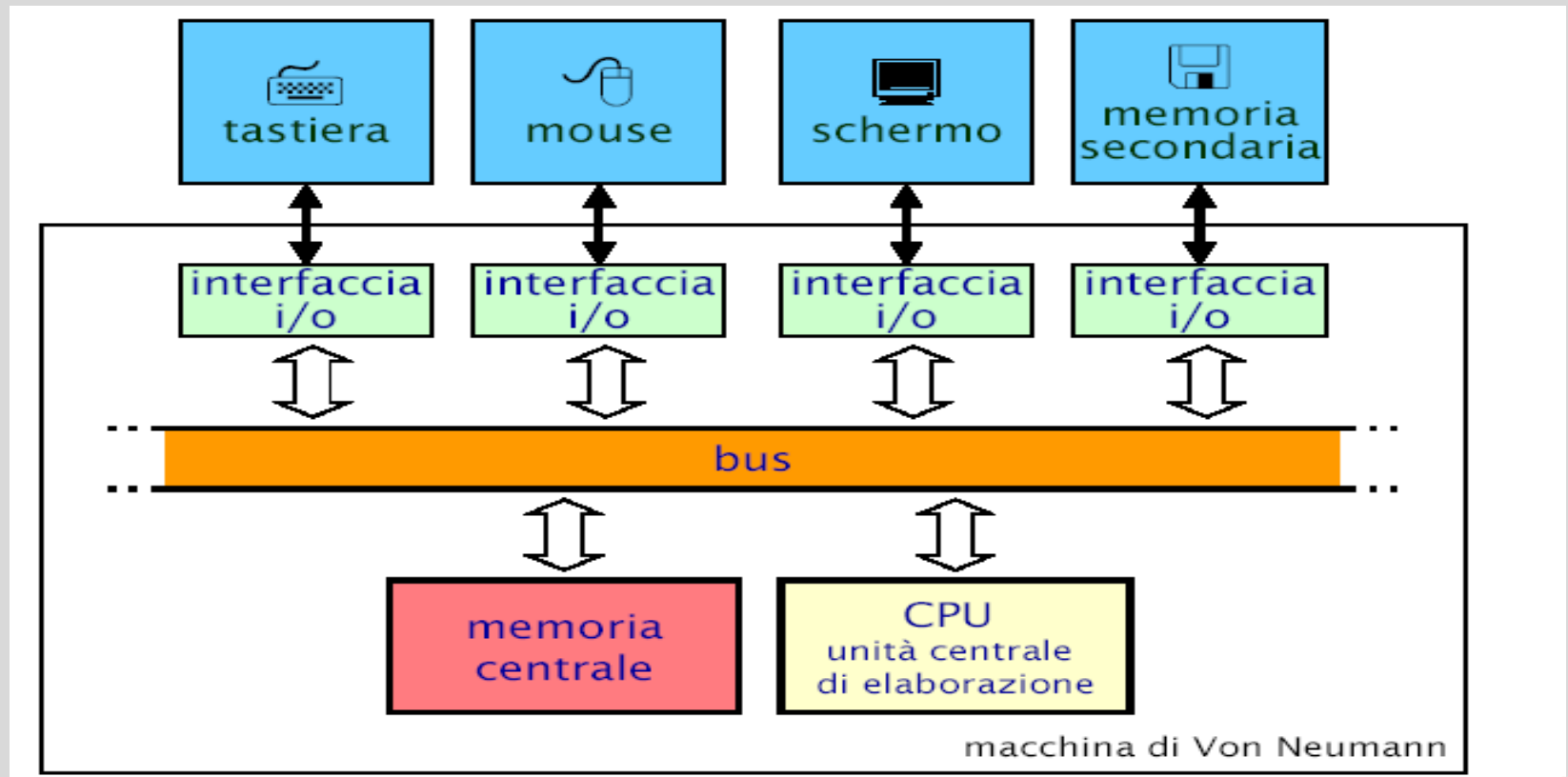
- Vari criteri
 - affidabilità
 - compatibilità
 - funzionalità
 - supporto e manutenzione
 - diffusione
 - costi
- Tipo di licenze
 - software proprietario
 - software libero
 - software senza licenza

Amministrazione di Sistema

Hardware

Hardware

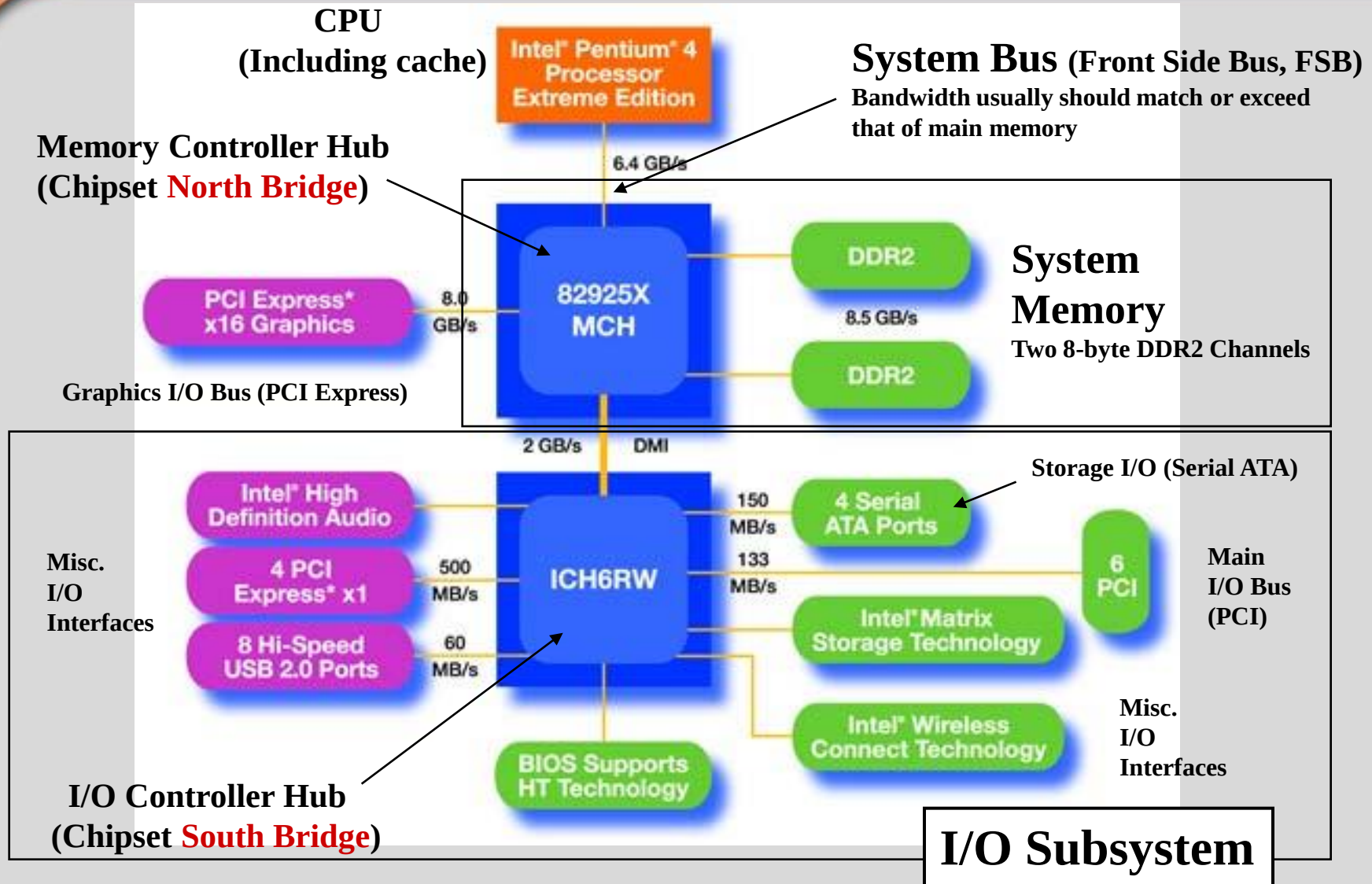
- Modello di riferimento
 - architettura di von Neumann
 - modificata in Architettura a Bus Multi-livello



Evoluzioni dell'hardware

- Più bus e più veloci
- Passaggio da UMA a NUMA
- xPU dedicati
- System on Chip (SoC)
- Memorie persistenti e Storage Class Memory
- SSD
- Risorse locali vs. risorse remote

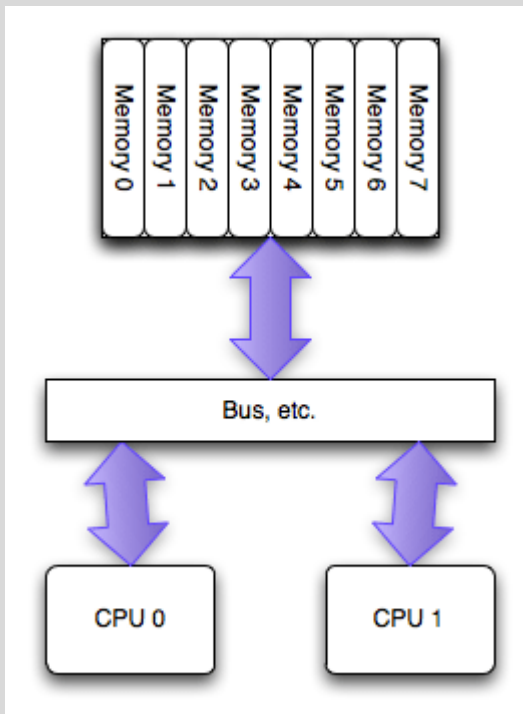
Intel Pentium 4 System Architecture



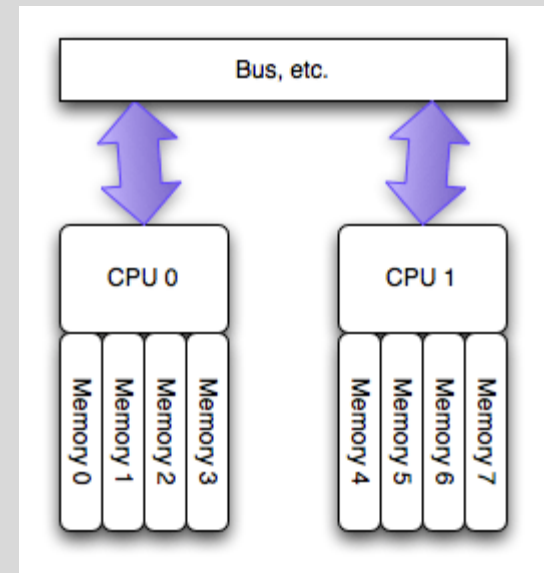
Source: The Intel 925 Chipset, 2004

NUMA vs. UMA

- Uniform Memory Access



- Non-Uniform Memory Access



Scelta dell'hardware

- Piattaforma di riferimento
- Continua crescita tecnologica
 - crescita non uguale tra le varie tecnologie (gap tecnologico)
 - “legge di Moore”
- Continua evoluzione tecnologica
 - processori Xeon e Opteron a 64 bit con fino a 64 Core
 - memorie veloci DDR4 (e DDR5)
 - PCI Express x4, x8, x16
 - Nuovi bus e nuovi modelli
 - Ethernet da 10Gbps a 100Gpbs
 - Trusted Computing Platform Alliance (TCPA)
- Marche e prodotti?

Prodotti di marca

- Meglio un server di marca o un assemblato?
 - Affidabilità di un server
 - Manutenzione
 - Contratti entro le 2 o 4 ore
 - Componenti disponibili per un certo periodo
 - Certificazione dell'hardware verso il sistema operativo o l'hypervisor
- Altri aspetti da considerare
 - Strumenti di gestione
 - Console remotizzabile
 - Design studiati per il funzionamento 24h x 365g
 - Flussi termici, soluzioni di hot-plug, ...

Caratteristiche dei processori

- Affidabilità (Reliability), stabilità (Stability), disponibilità (Availability) e ripristinabilità (Recoverability)
 - Nei Mainframe vi sono soluzioni specifiche
- Scalabilità (Scalability)
 - Architetture multi-socket e multi-core
 - Con più nodi NUMA
- Prestazioni (Performance)
 - Hyper-Threading
 - Turbo Boost
 - xPU
- Manutenibilità (Manageability) e semplicità
 - Verso soluzioni SoC
- Sicurezza (Security)
 - Problema di bug hardware

Scelta del processore

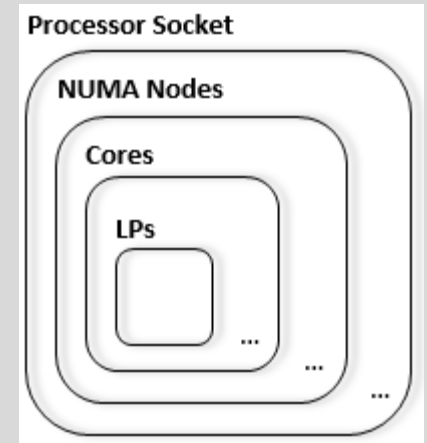
- Funzioni tipiche
 - SMP (Symmetric Multiprocessing)
 - motherboard per server bi/quadri-processore
 - Multi Core (attualmente fino a 64 core ma 122 in roadmap)
 - <http://ark.intel.com/it>
 - HTT (Hyper-Threading Technology)
 - Cache L1, L2, L3
 - Fino a quasi 100 MB
- Convenienza? Differenze tra i modelli?
 - Top della gamma > 15K€
 - Costi hardware e costi di licenza?
 - Costo per socket o per core?
- Architetture
 - Processori a ~~32-bit~~ o 64 bit
 - Per maggiori informazioni
 - http://it.wikibooks.org/wiki/Architetture_dei_processori

Processori a 64 bit

- Piattaforme a 64 bit
 - X64 (2000 AMD, 2005 Intel)
 - AMD64 (x86-64), Intel EM64T
 - Modalità legacy 32 bit, compatibility, full 64 bit
 - ~~Itanium (2001)~~
 - ~~Intel IA-64~~
- Applicazioni solo a 64 bit
 - Exchange 2007
- Hypervisor solo a 64 bit
- Sistemi Operativi server solo a 64 bit
 - Microsoft Windows Server 2008 R2
 - Microsoft SBS 2008
 - Distribuzioni Linux recenti

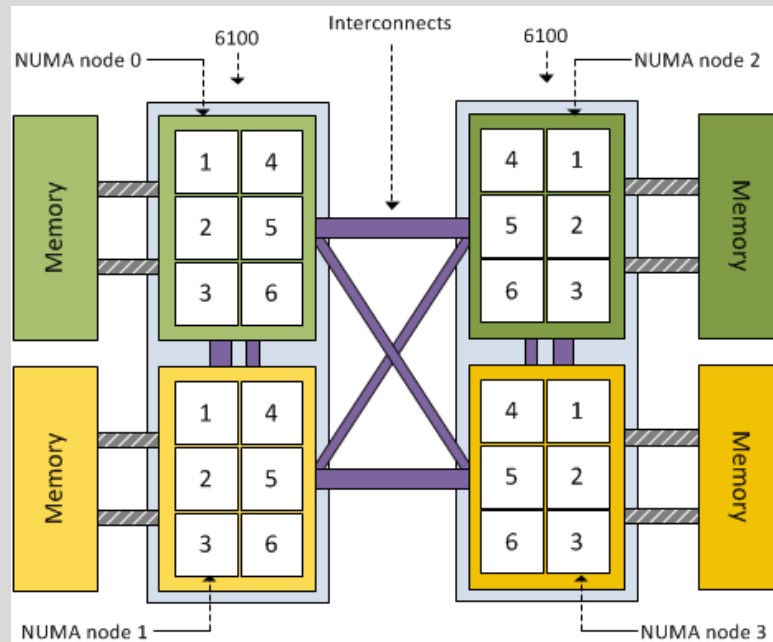
Terminologia delle CPU

- Socket
 - La CPU fisica
- Nodo NUMA
 - Parte della CPU con accesso alla RAM condiviso
- Core
 - Unità di elaborazione
- Processore logico
 - Unità logica
 - Hyper-Threading
 - <https://it.wikipedia.org/wiki/Hyper-Threading>
- NUMA vs. UMA



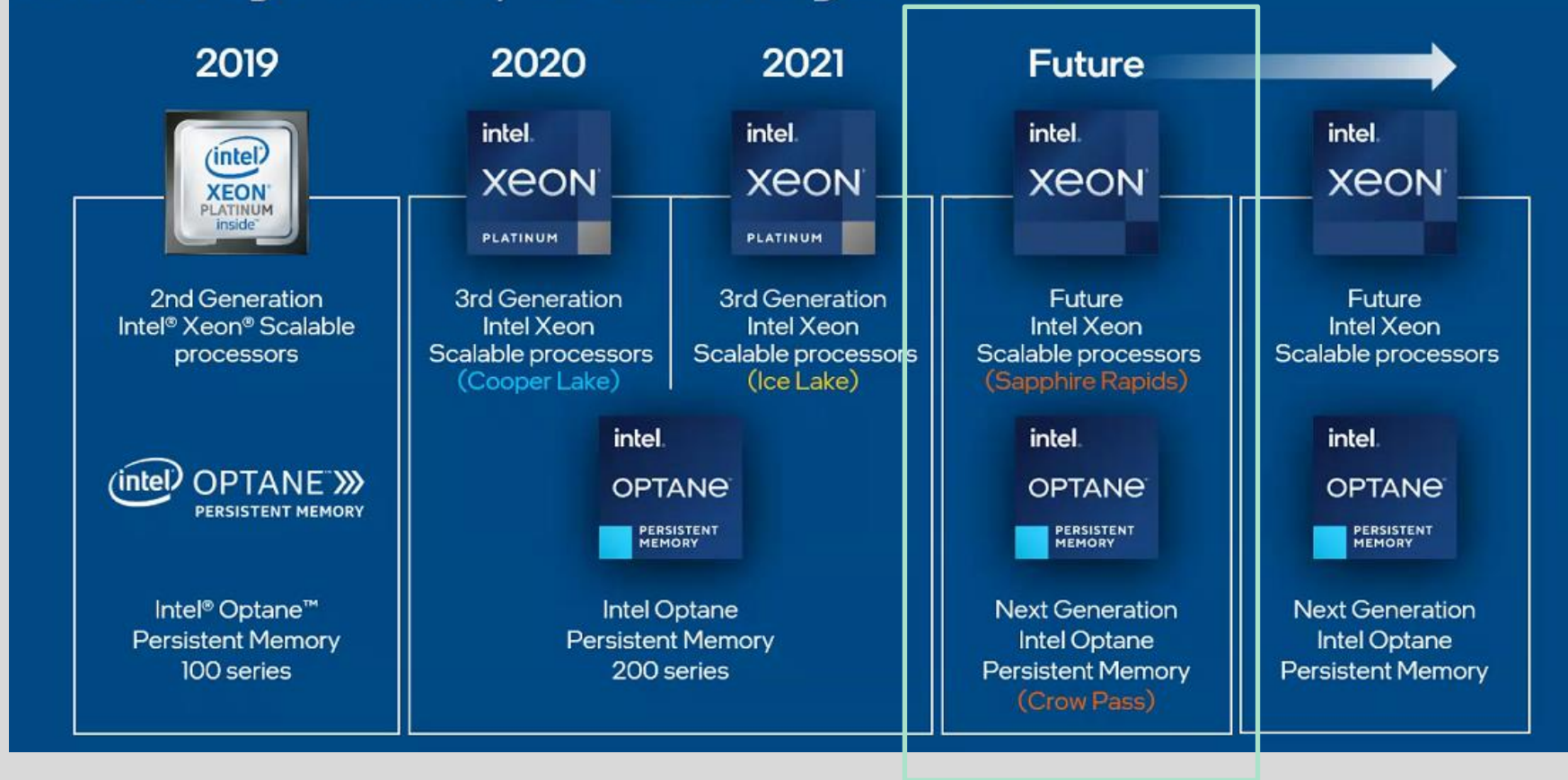
Architettura delle CPU

- Processori moderni
 - Internamente UMA di multicore
 - Esternamente NUMA con memoria dedicata
 - Più core che frequenza

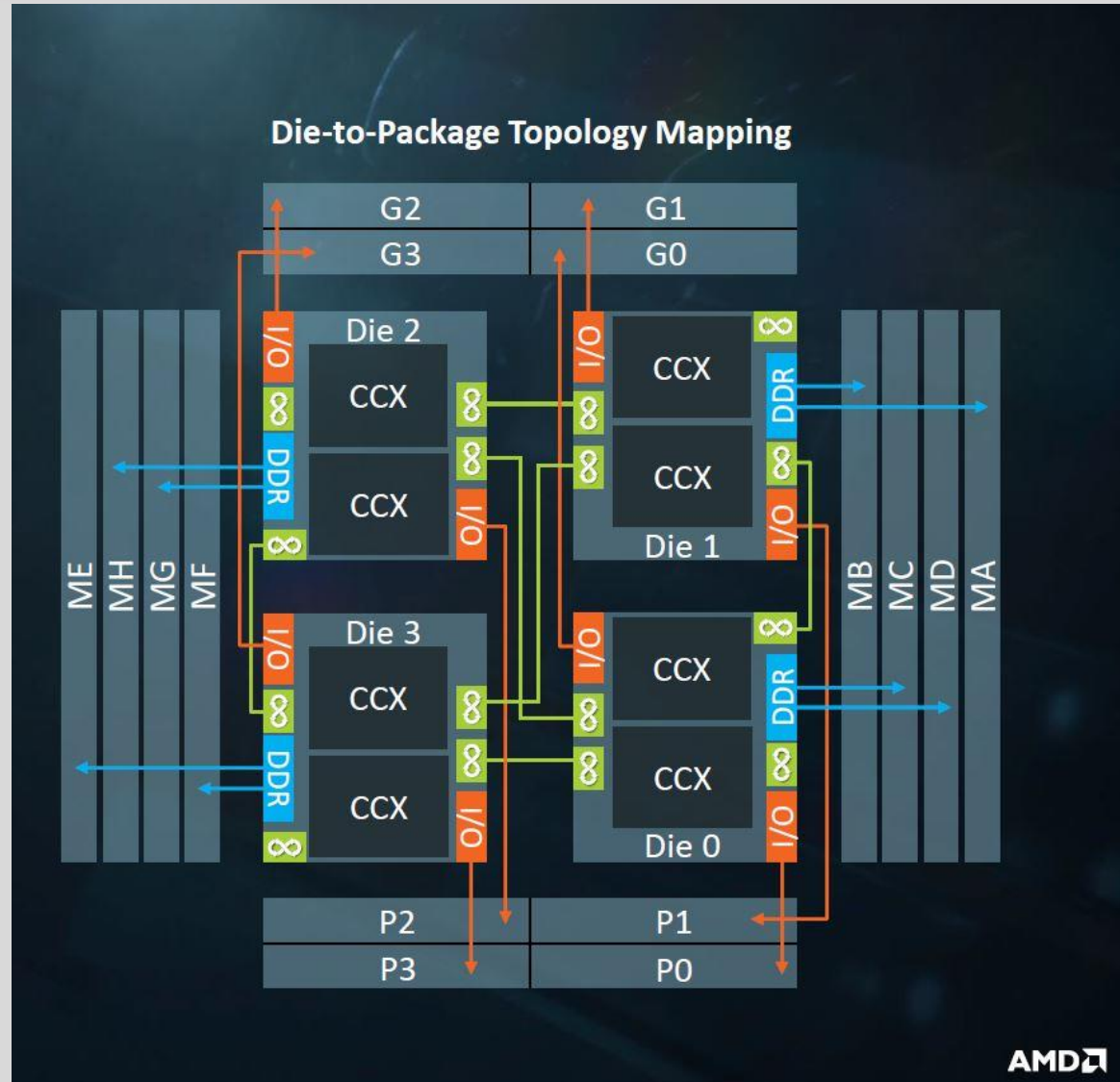


Intel Xeon

A Strong Memory and Storage Future



AMD EPYC



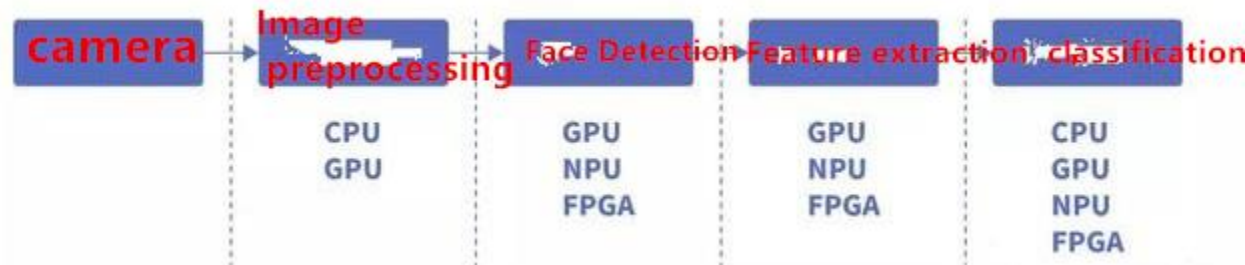
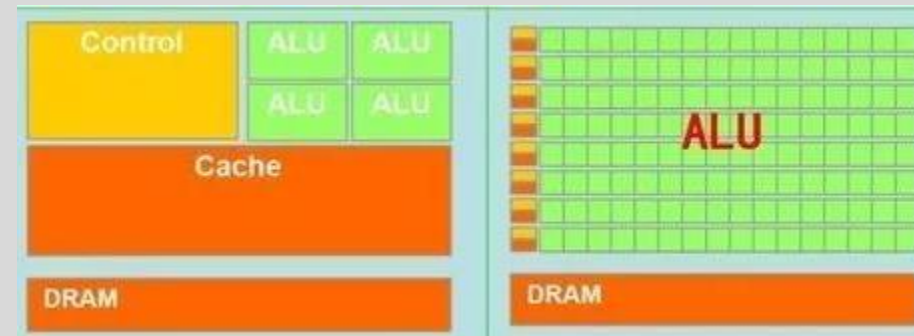
Processori e sicurezza

- Le CPU sono le nuove superfici di attacco
- Diversi bug
 - 2018 anno chiave
 - Spectre, Meltdown
 - L1TF, Foreshadow, ZombieLoad, ...
 - Ma anche prima del 2018
 - Ad esempio: Intel Management Engine (ME) bug
- Mitigare i problemi
 - Nuovo microcode
 - SO assisted patch
 - Hypervisor assisted patch
- Impatto delle varie fix sulle CPU
 - Intel 9-th Gen presentano fix hardware



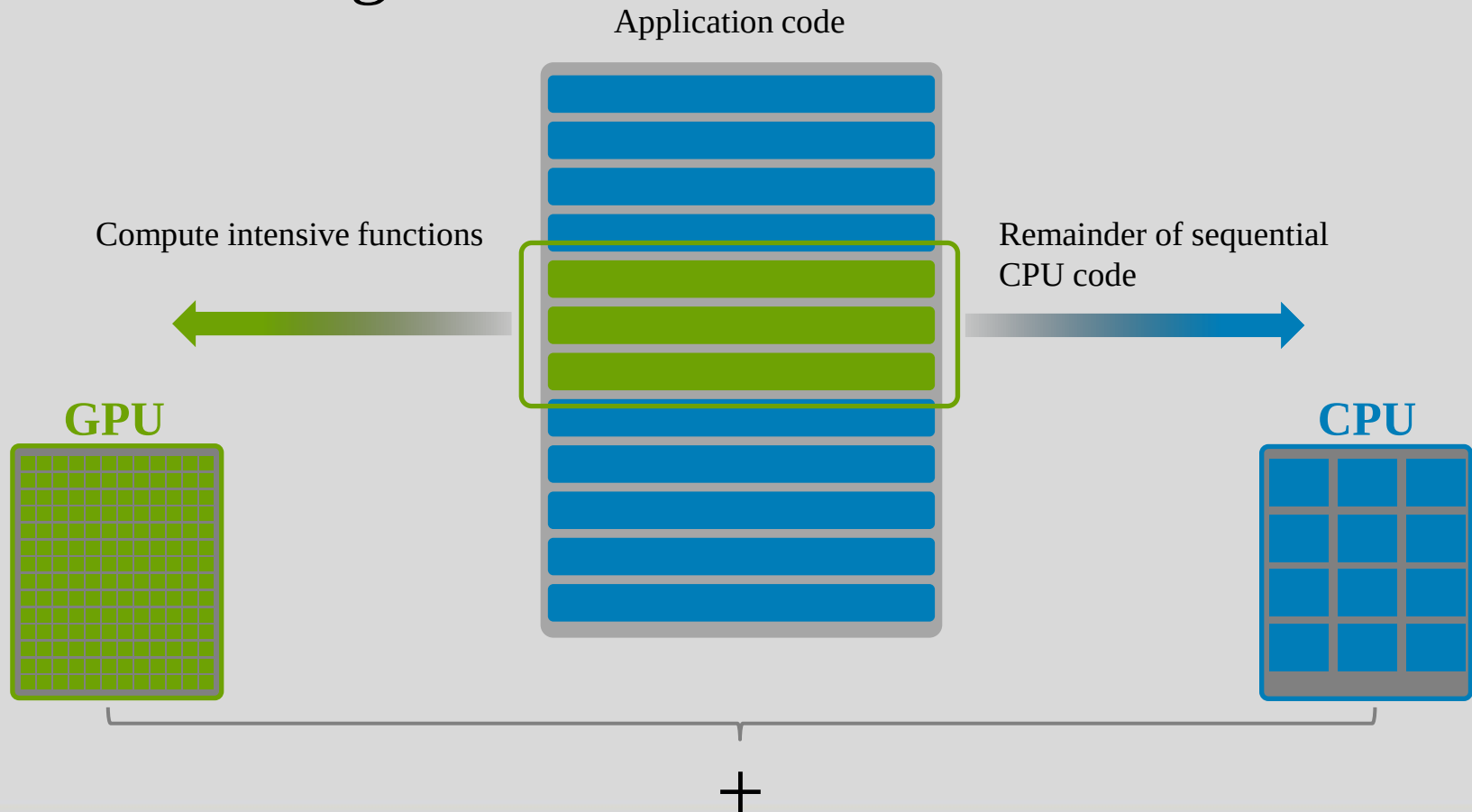
Evoluzione del calcolo

- CPU
 - Coprocessore matematico
 - Istruzioni SSE, SSE2, ...
- (GP)GPU
 - CUDA (Compute Unified Device Architecture)
 - OpenCL
 - ...
- FPGA
- NPU (Neural Processing Unit)
 - acceleratore IA

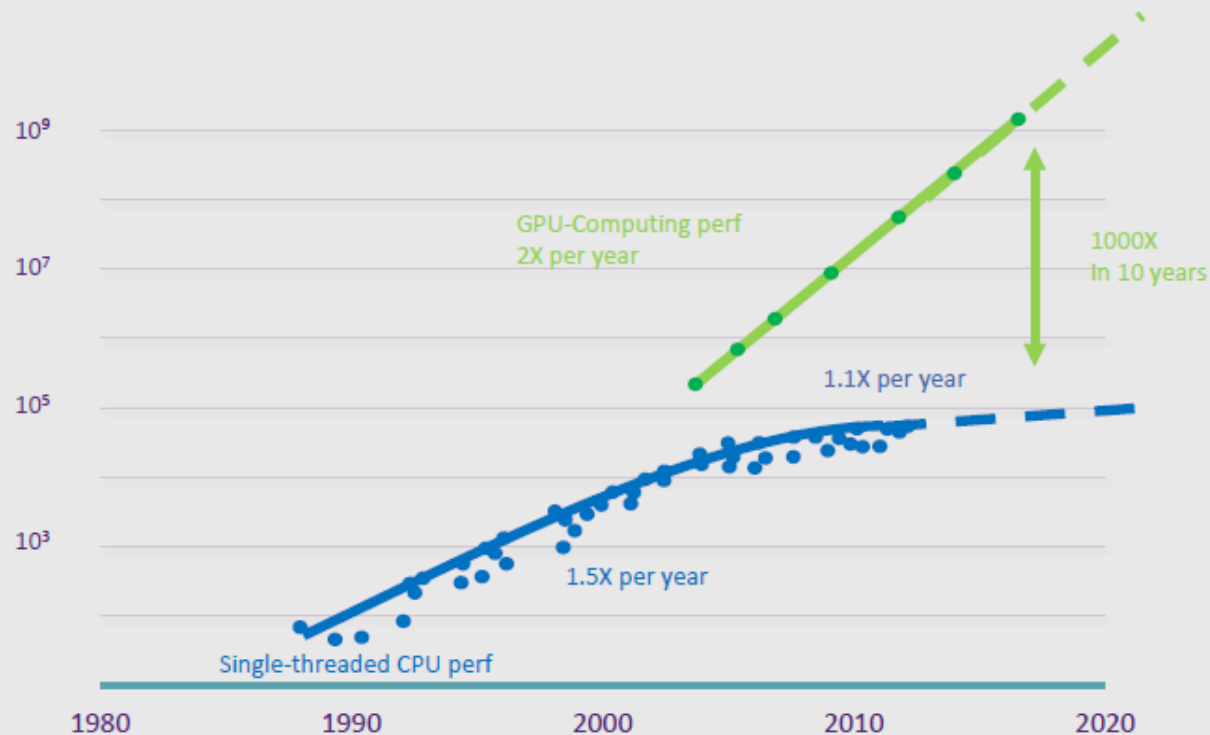


GPGPU

- General-Purpose Computing on Graphics Processing Units



The new General Processing Unit

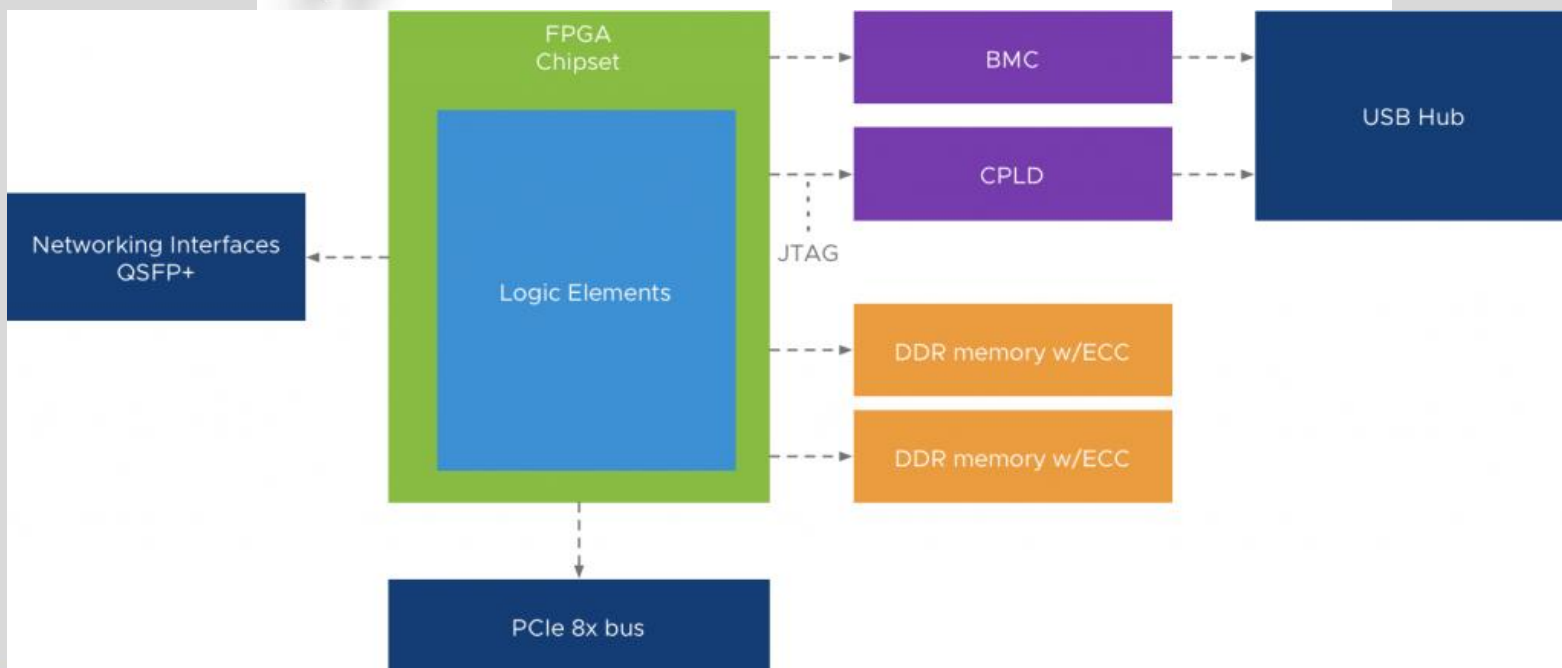


FPGA

- Field-Programmable Gate Array
 - integrated circuit that can be programmed in the field after manufacture
 - in contrast to processors that you find in your PC, programming an FPGA rewires the chip itself to implement your functionality rather than run a software application
 - they can be used to accelerate code that runs well when executed in parallel, for example deep machine learning



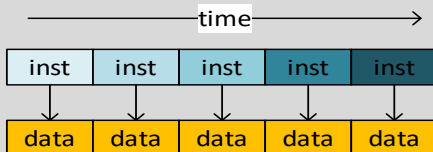
FPGA



CPU vs GPU vs FPGA

CPU - SISD

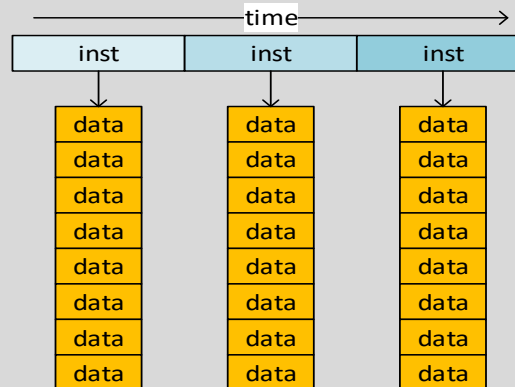
- Applicable to any programmable environment and language.
 - Non-vectorized integer applications
- Remains the core center of computation



GPU - SIMD

- Excels at vectored floating point
- Requires large data to be effective
- Hurt by branches or exceptions – “if” statements

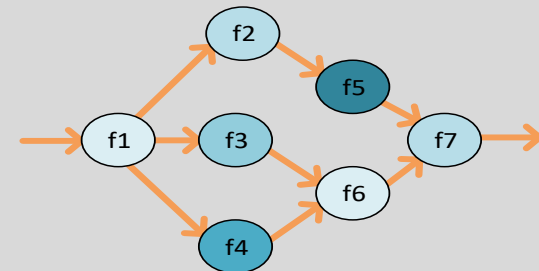
- Remains focused on a subset of high-performance problems
- Not going away; limited breadth



FPGA - MIMD

- Many independent instructions can operate in parallel and with either small amounts or large amounts of data
- Excellent streaming with IO devices

- Greater breadth than GPUs
- Expect higher prevalence for enterprise applications than GPUs over time



Server con GPU

- Dell
 - Dell PowerEdge XE9680
 - <https://www.dell.com/en-us/blog/dell-generative-ai-open-ecosystem-with-amd-instinct-accelerators/>



- HPE
 - Cray XD670
 - <https://www.hpe.com/it/it/hpe-cray-xd670.html>

xPU

- Non solo CPU, GPU e FPGA
- Ma anche:
 - ASIC - Application Specific Integrated Circuit
 - TPU - Tensor Processor Unit
 - NPU - Neural Processor Unit
 - IPU – Intelligent Processing Unit
 - Developed by Graphcore the IPU is specifically designed for AI/ML
 - DPU – Data Processing Unit
 - Processing at the edge, in the network interface cards
 - Leading providers are NVIDIA and Intel

xPU a confronto



High Computing Efficiency

(Performance of a high-end DSP/GPU)



Power Efficiency

(Sub 30W chip typ.)



Easily Programmable

(C/C++/Open CL; Linux; POSIX; RTOS)



Concurrent Execution of Dozens of Heterogeneous Critical Tasks



Low Latency / Deterministic / Real-Time

(High speed I/O – RDMA type of architecture)



Performance Scalability

(within die; die-to-die; many dice)

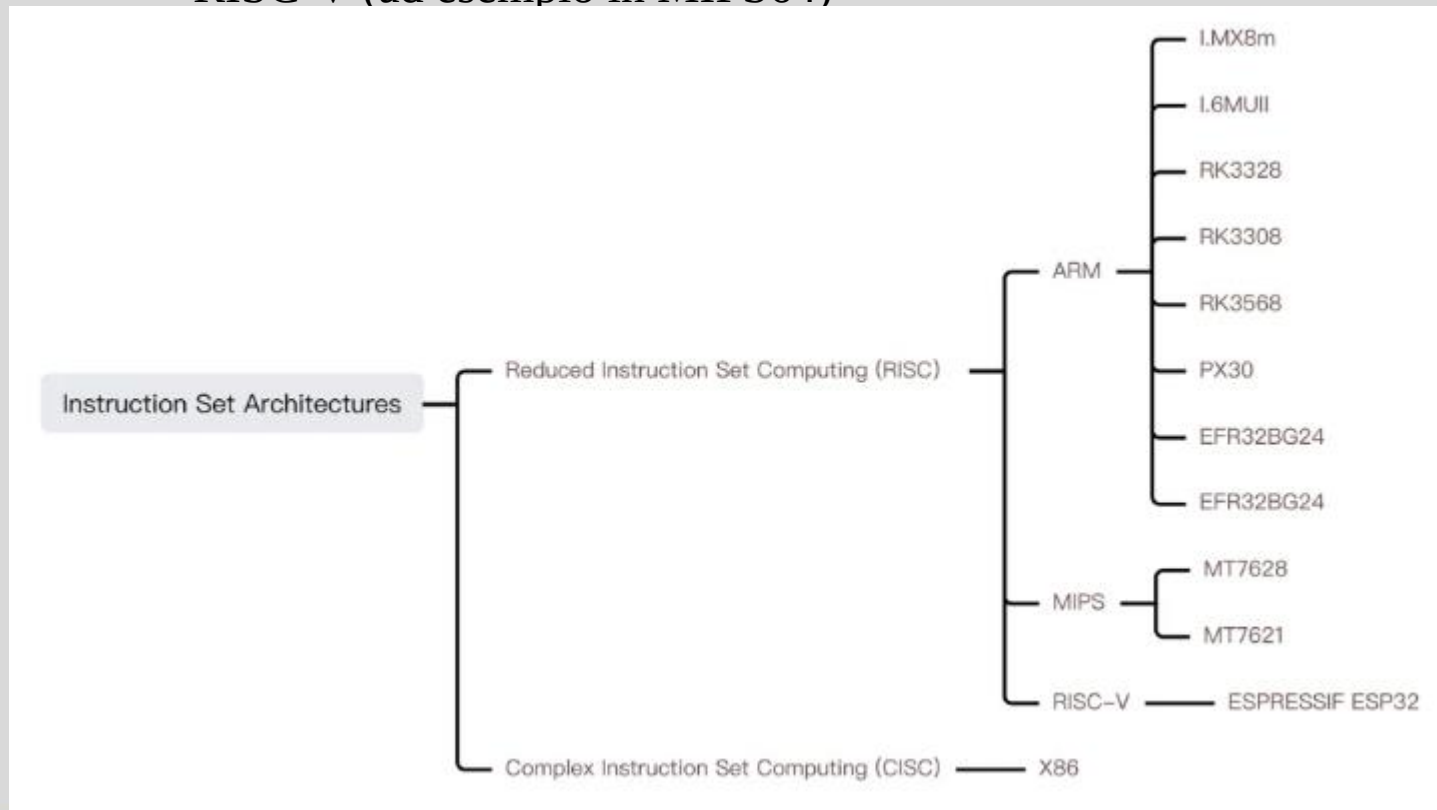
DPU	FPGA	GPU	x86
✓	✓	✓	
✓		✗	✗
✓	✗		✓
✓	✓	✗	✗
✓	✓	✗	✗
✓			

Alternative a x64?

- Alternative basate su processore ARM (Advanced RISC Machines)
 - Datacenter
 - Ampere Computing
 - ARM Neoverse N1 System
 - Near-edge
 - SolidRun
 - NVIDIA Jetson AGX Xavier
 - Edge
 - Raspberry Pi
 - NVIDIA Jetson Xavier NX
 - LS1046A-based NXP
 - Socionext SynQuacer
 - PINE64 Quartz64 Model A

Alternative ad ARM?

- Altri processori RISC
 - MIPS (Microprocessor without Interlocked Pipelined Stages)
 - OpenSource Instruction Set Architecture (ISA)
 - RISC-V (ad esempio in MIPS64)



CISC vs. RISC

- Architetture per micro-processorii
 - Complex Instruction Set Computer (CISC)
 - Reduced Instruction Set Computer (RISC)
 - Architetture ibride

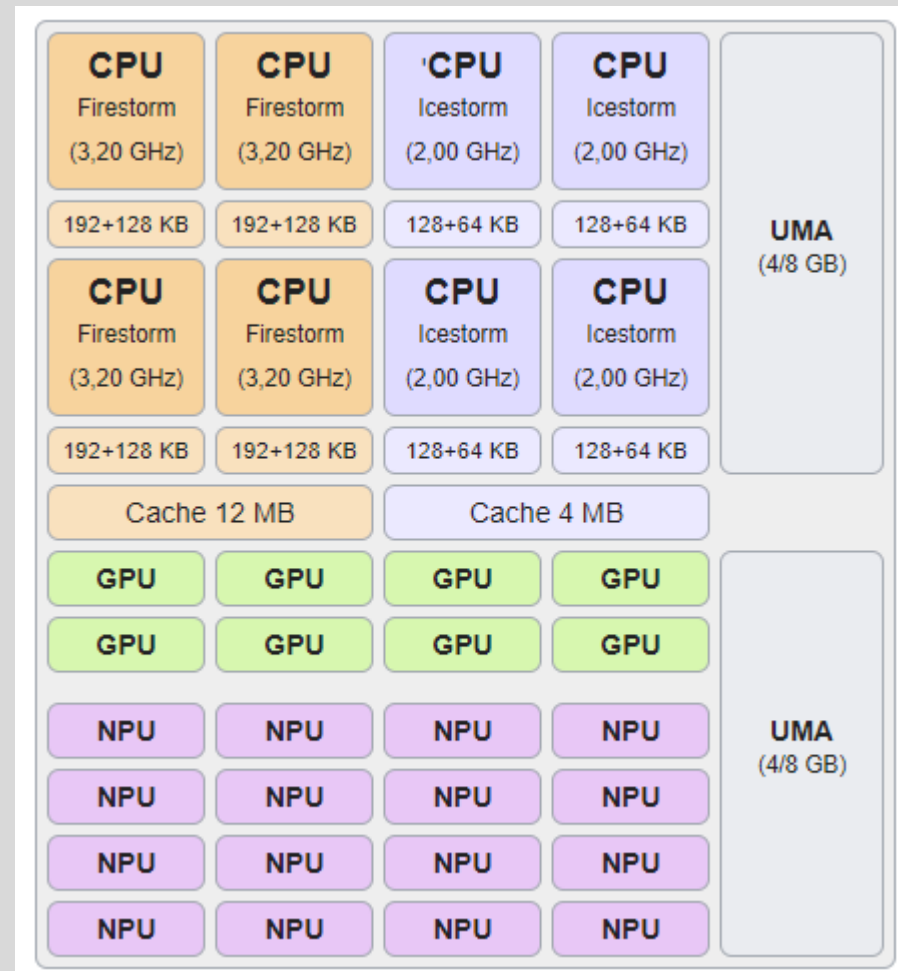
- Differenze?

<https://microcontrollerslab.com/difference-between-risc-and-cisc/>

CISC	RISC
Memory unit is present to implement the instructions	There is no memory unit and registers store data
It is microprogramming unit	It has a complex design of compiler
Its compiler design is easy	Compiler design is complex
Its calculations are slower yet precise	Perform mathematical calculations faster
Their decoding is difficult	Decoding of its instructions is easier
Instructions are complex so it takes time in execution	It is faster as its instructions are simple
External memory mandatory requirement	No external memory requirement
Pipelining is difficult to implement	Pipelining is easy to implement
Their processors often stall	There is no stalling normally
Code expansion is easier	The expansion of code can be an issue
Utilize more disk space	Consumes less disk space
Its examples include: include VAX, PDP-11, Motorola 68k and your desktop PCs	Its examples include: DEC Alpha, ARC, AMD 29k, Atmel AVR, Intel i860, Blackfin, i960, Motorola 88000, MIPS, PA-RISC, Power, SPARC, SuperH, and ARM

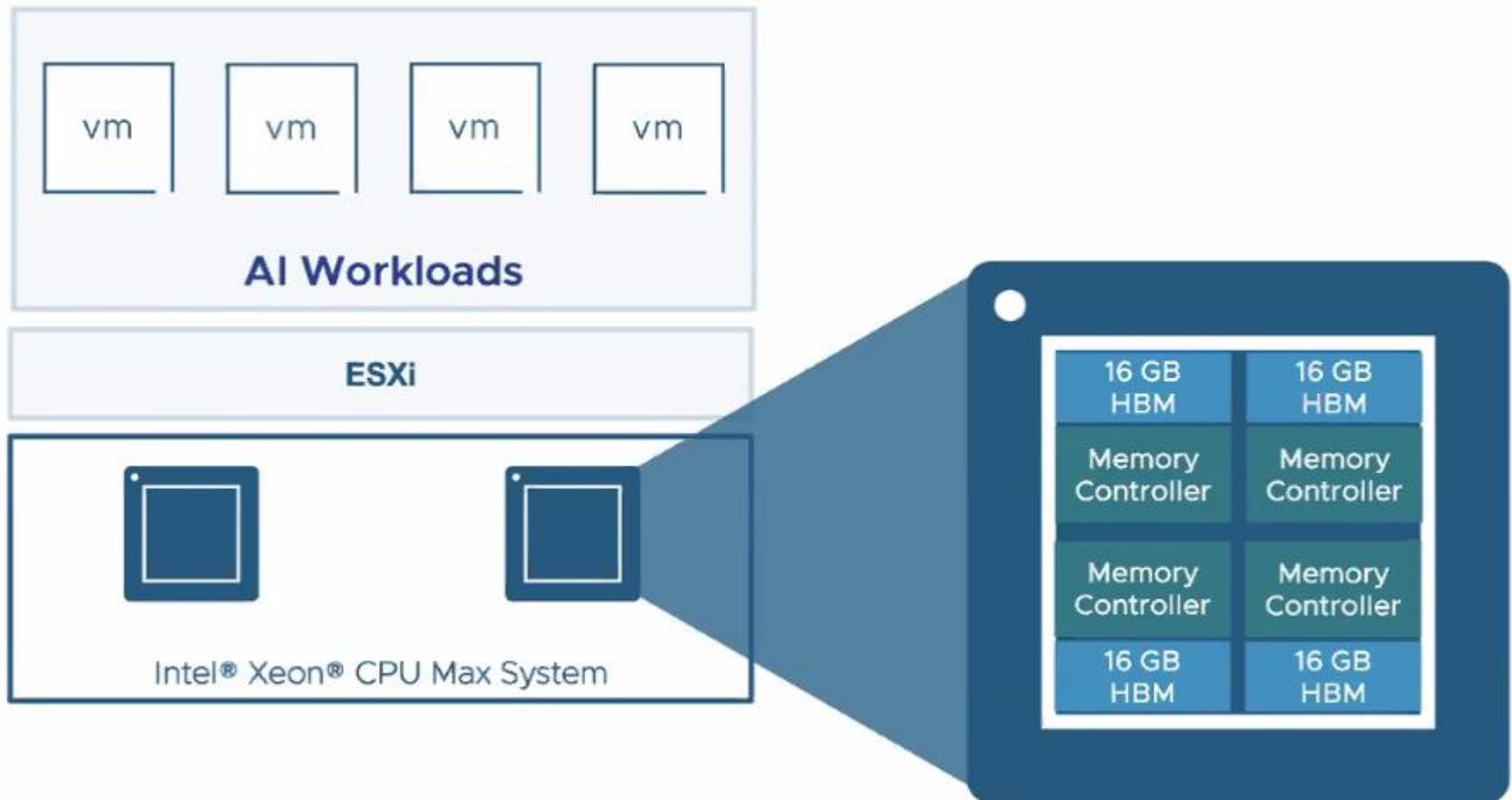
SoC

- Processori Apple
 - Apple M1
 - System on a chip (SoC) a 64-bit
 - RISC
 - Introdotto nel 2020
 - Nuova famiglia Mac



SoC

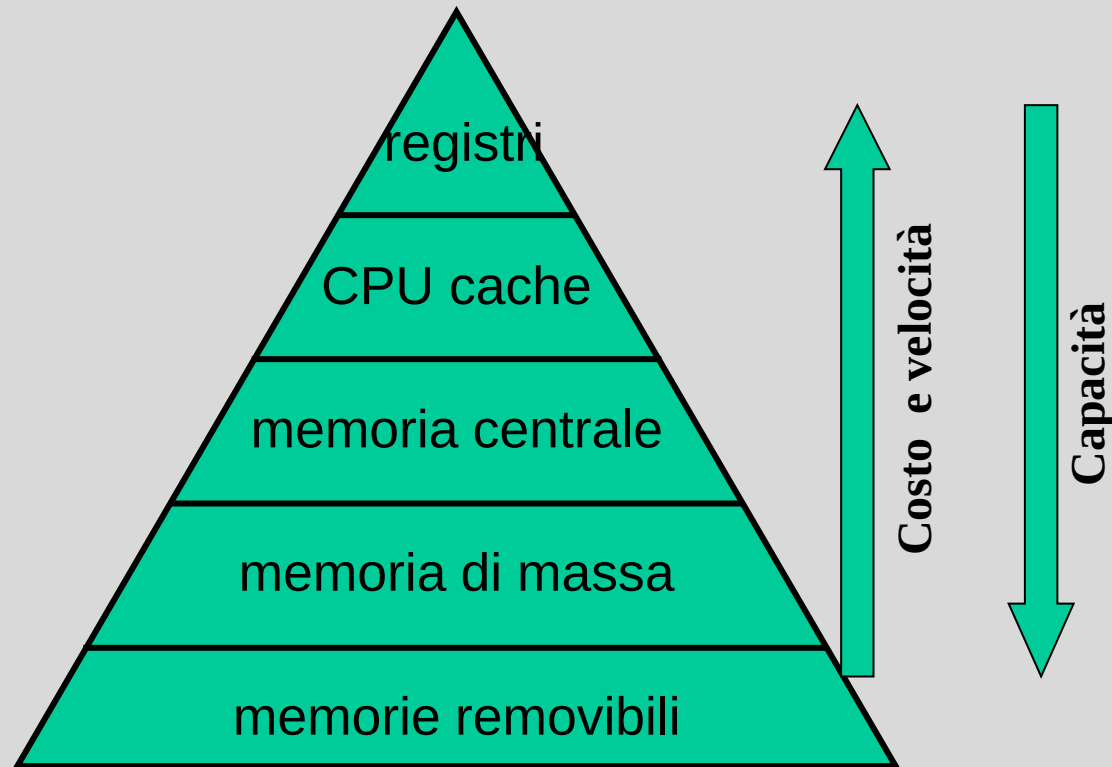
- Intel Xeon Max



Memorie

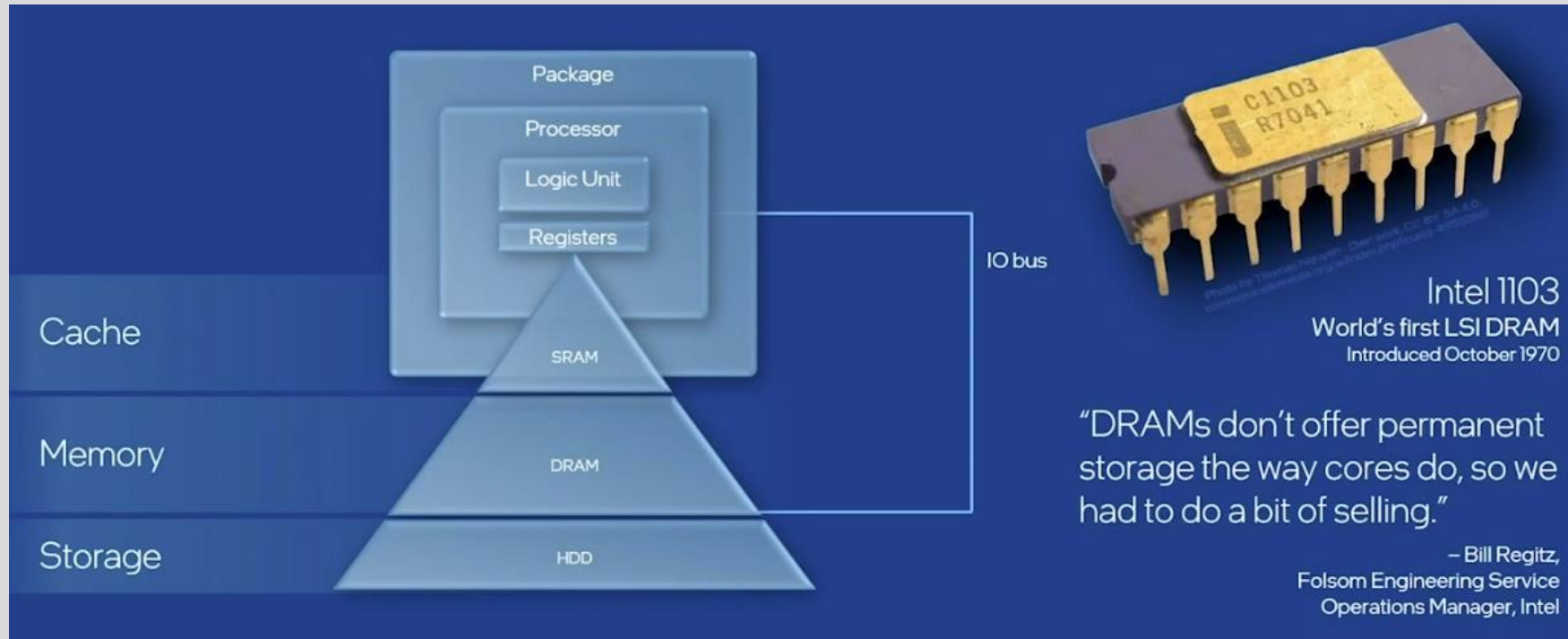
- Volatili
 - RAM
 - SDRAM, DDR, GDDR, HBM
 - SRAM
- Non volatili
 - ROM
 - Mask ROM, PROM, EPROM, EEPROM, Flash memory
 - NVRAM, ReRAM
 - Magnetic
 - Optical

Memory/storage hierarchy



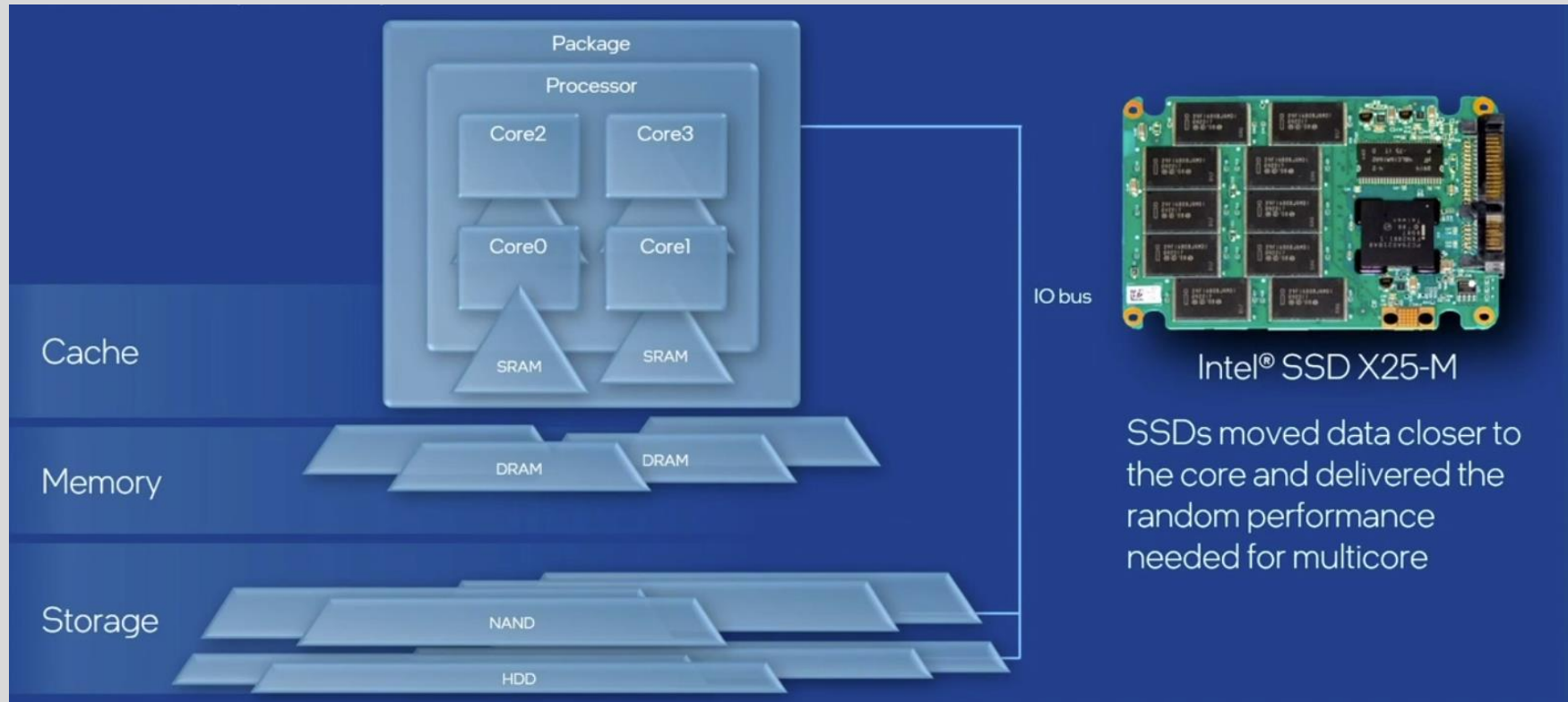
Hierarchy evolution

Gli albori



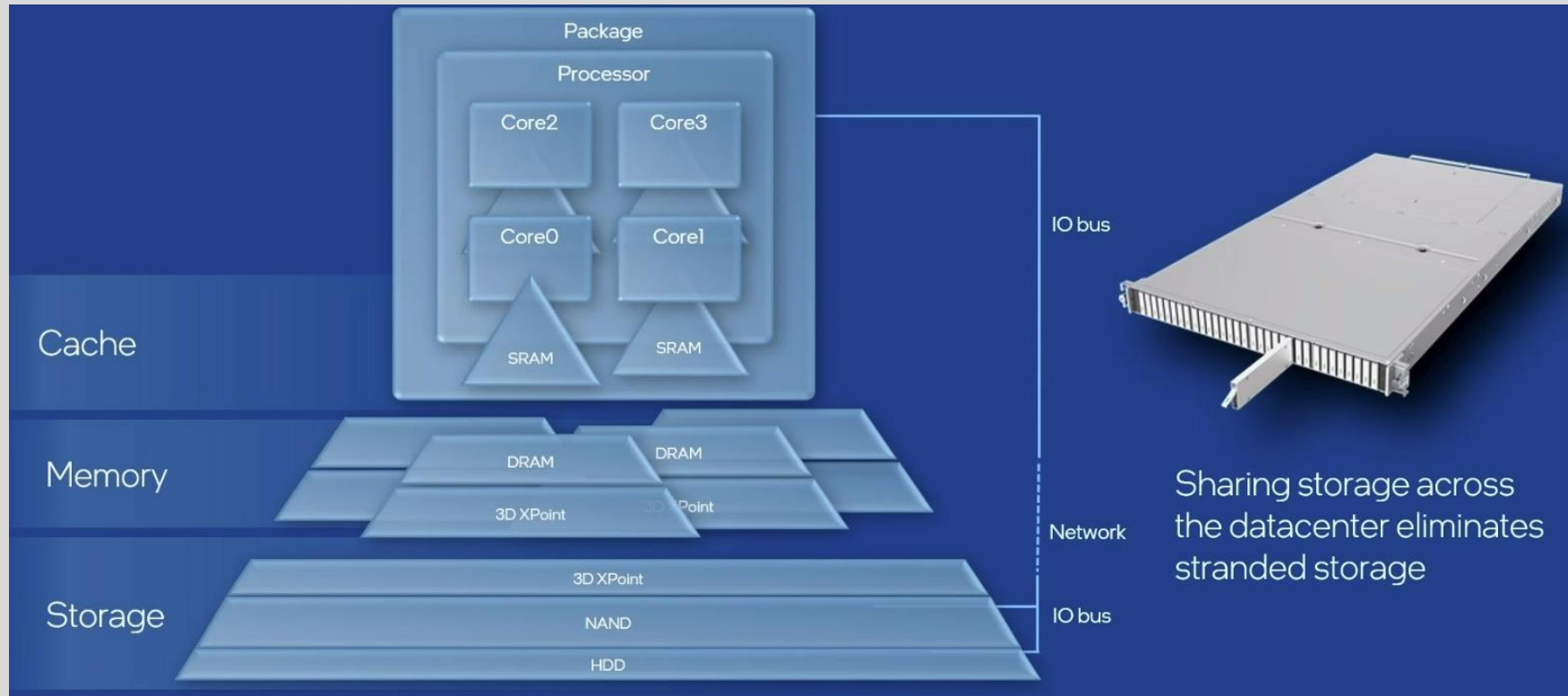
Hierarchy evolution

Multi-core (2000 in poi)



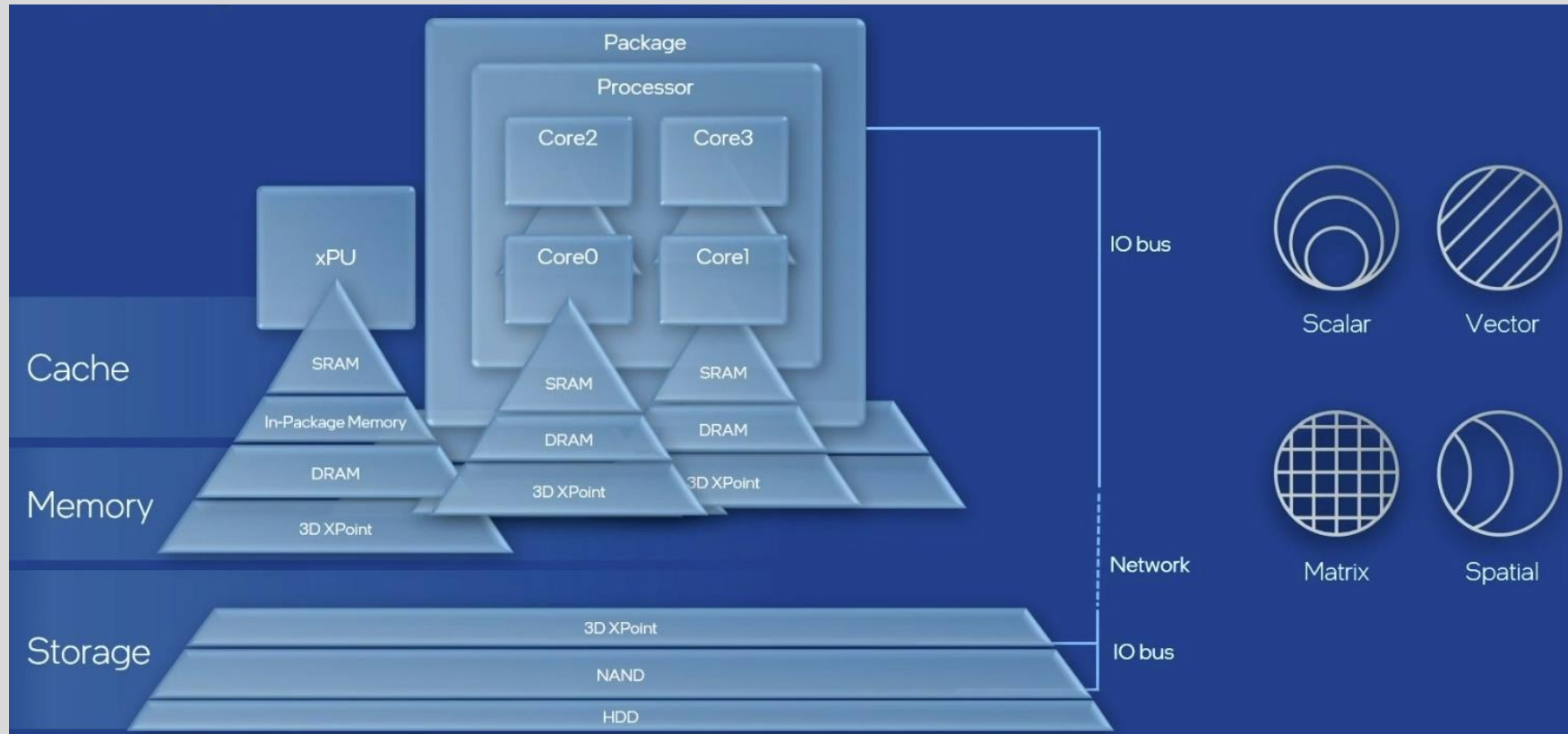
Memory/storage hierarchy evolution

Cloud computing (2010 in poi)



Hierarchy evolution

Architecture era (2020 in poi)

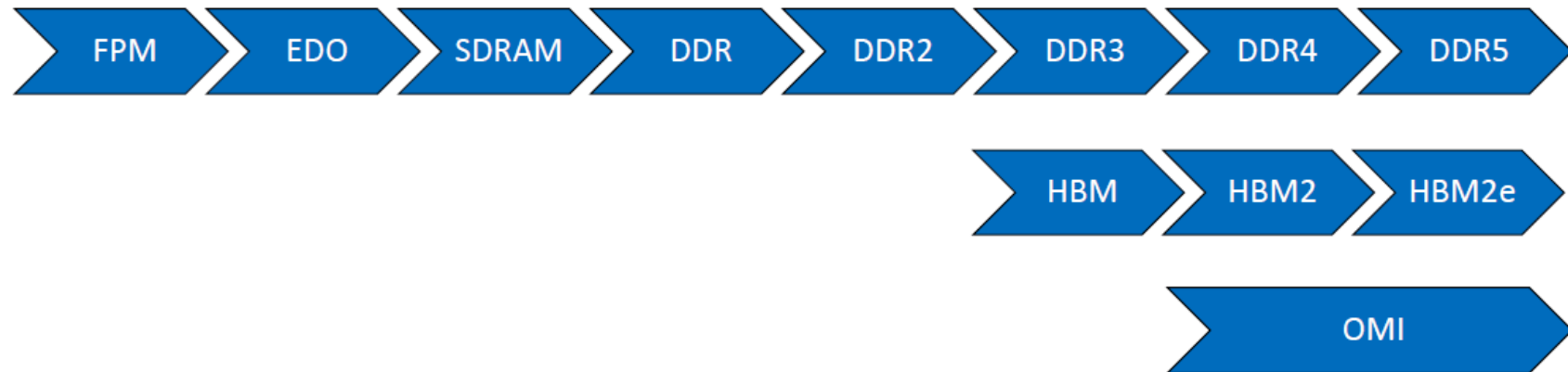


Caratteristiche delle RAM

- Affidabilità (Reliability), stabilità (Stability), disponibilità (Availability) e ripristinabilità (Recoverability)
 - Soluzioni ECC
 - Soluzioni di ridondanza hardware con hot-swap
- Scalabilità (Scalability)
 - Più canali per CPU (tipicamente 6)
 - Più DIMM per canale (tipicamente 2)
 - Verso memorie persistenti
- Prestazioni (Performance)
 - Configurazioni ottimali
- Manutenibilità (Manageability) e semplicità
 - Formati standard
- Sicurezza (Security)
 - Definizione di enclave di sicurezza nei nuovi processori Intel e AMD

Scelta delle memorie RAM (1)

- Velocità delle memorie
 - DDR SDRAM
 - Server attuali con 12 slot per socket
 - Attualmente **DDR4 e DDR5**
 - *Alternative?*
 - ~~Rambus XDR (e XDR2)~~
 - OMI o HBM2E

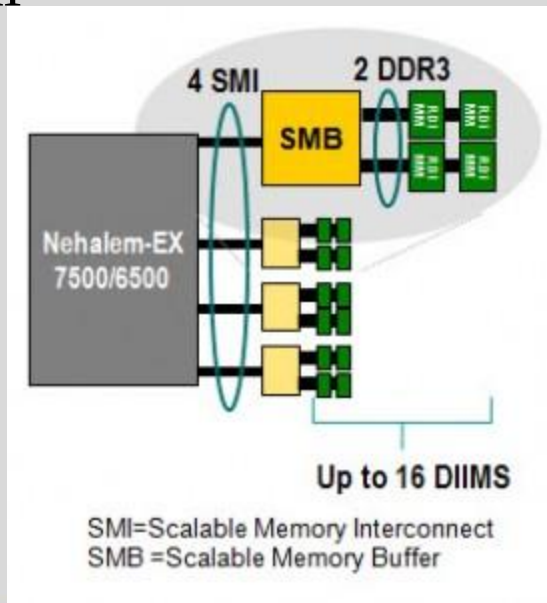
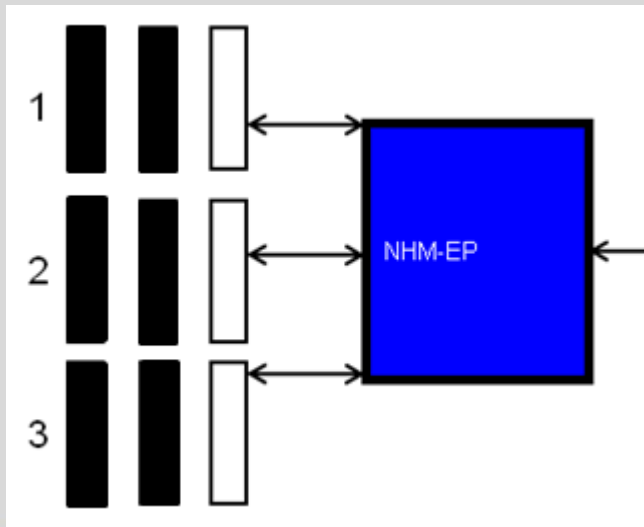


Scelta delle memorie RAM (2)

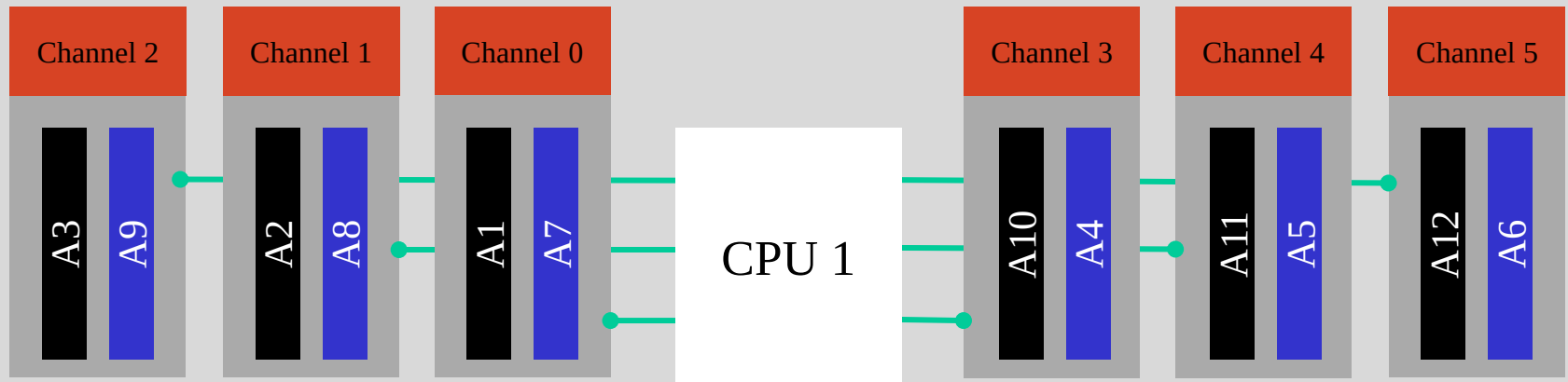
- Problema dei prestazioni
 - Architetture NUMA
 - Cache L1,L2,L3 lato CPU
- Problema di affidabilità
 - Memorie con ECC (Error Correcting Codes)
 - Solo per certi modelli di server di fascia alta:
 - configurazione dei banchi in modalità “mirror”
 - hot-swap dei banchi
- Problema della volatilità?
- Quantità massima installabile
 - anche 3TB di RAM (o più)
 - motherboard con più di 4 GB installabili
 - problemi con il Sistema Operativo?!

Considerazioni sulla RAM

- UDIMM Versus RDIMM
- Come “popolare” i banchi RAM?
 - <http://www.delltechcenter.com/page/04-08-2009+-+Nehalem+and+Memory+Configurations>
- Come ottimizzare i costi?
 - il costo al GB dipende dal tipo di banchi



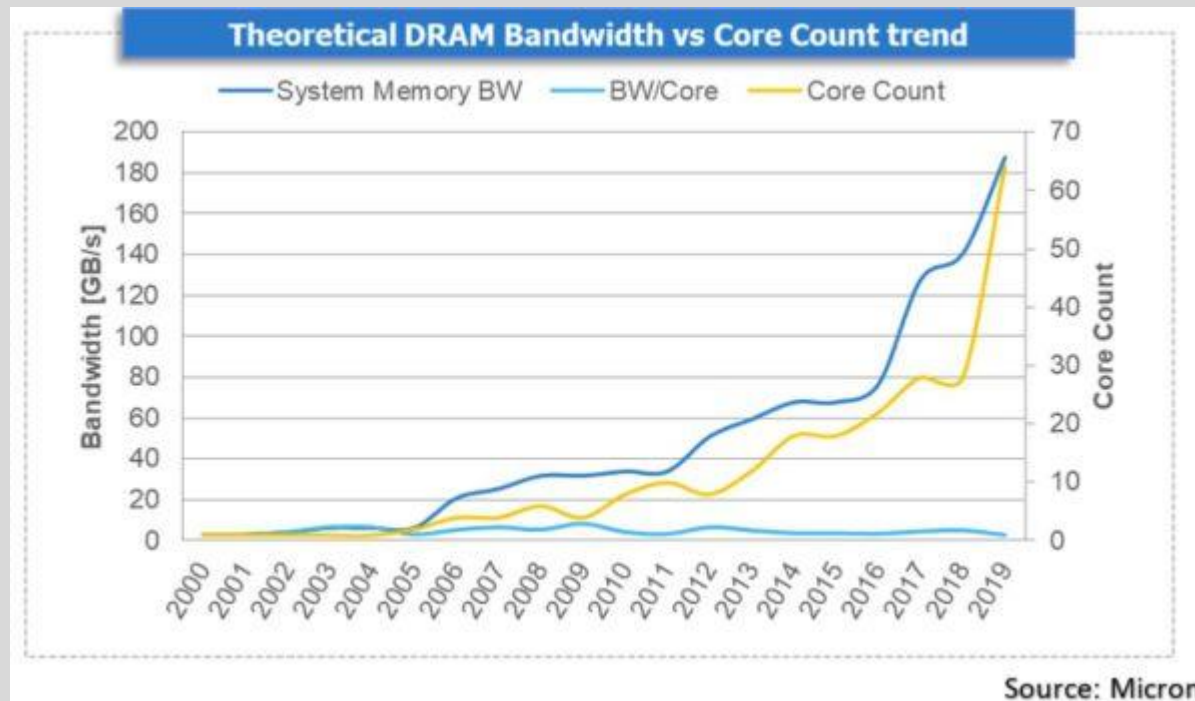
Skylake memory channels



6 memory channels x 2 slots per channel = 12
DIMMs per CPU

RAM e Core

- Troppi core in un nodo NUMA possono saturare la banda delle memorie



- le memorie DDR5 dovrebbero raggiungere un data rate massimo per pin di 6400 MT/s

RAM e Workload

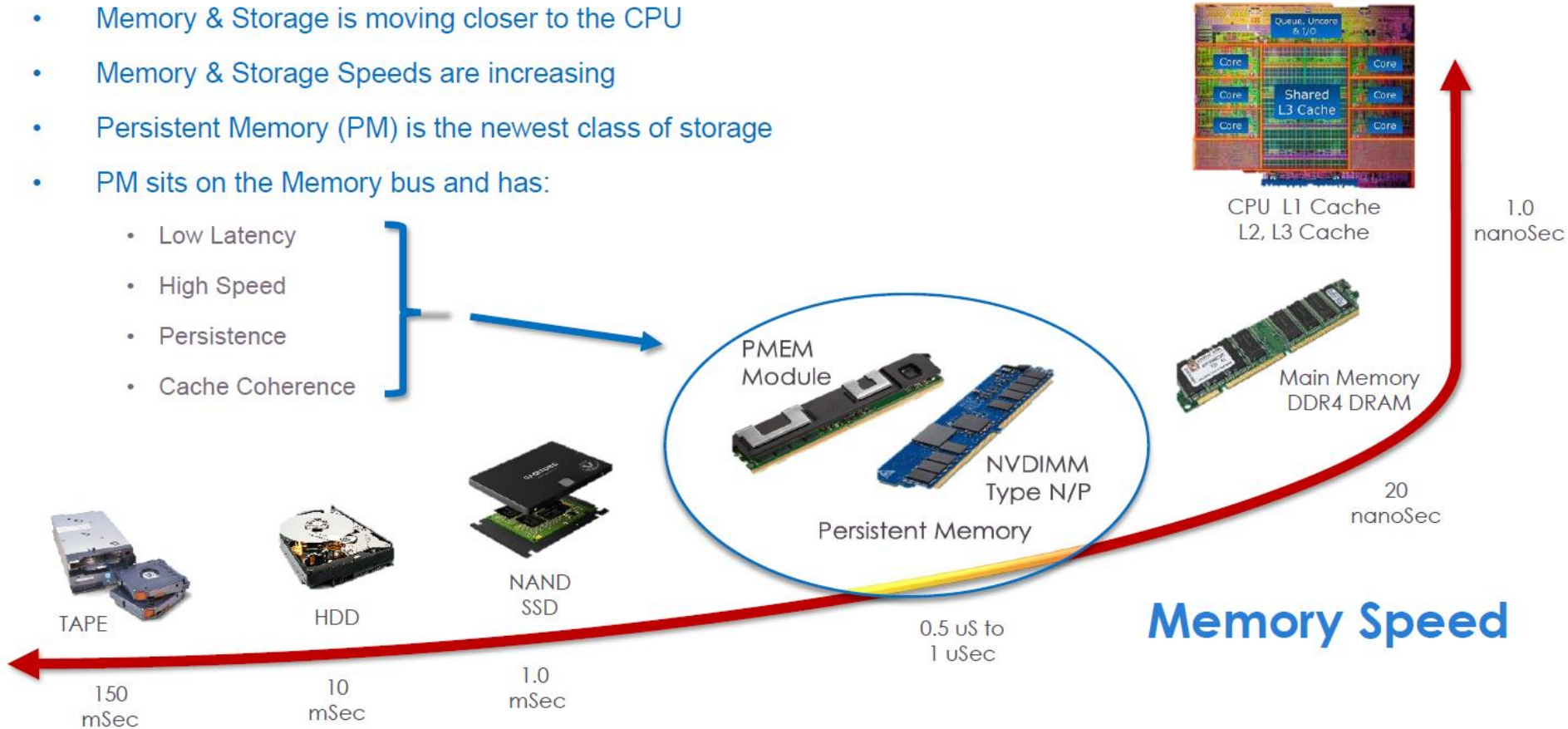
- Storicamente i processi lavorano su storage primario e secondario
 - I dati da storage secondario devono essere caricati su quello primario per diventare accessibili alla CPU
- DBMS che gestiscono i dati nella memoria primario
 - In-memory database (IMDB)
 - Main memory database system (MMDB)
 - Memory-resident database
- Permettono velocità maggiori
 - Ma anche di implementare in modo efficiente modelli diversi dal relazionale come il reticolare, gerarchico o ad oggetti

Oltre le RAM?

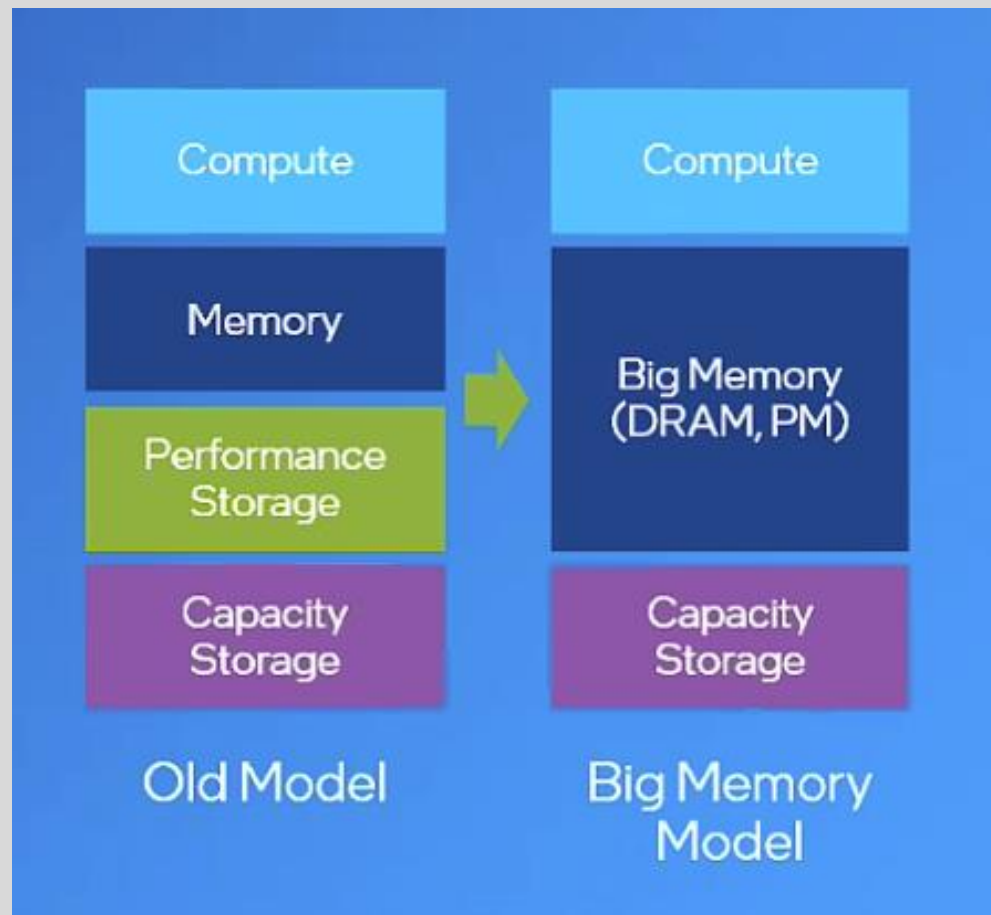
- Flash al posto della RAM?
 - Velocità / Costi ?
- Standard NVDIMM
 - NVDIMM-F (introdotte nel 2014)
 - DIMM with flash storage
 - NVDIMM-N
 - DIMM with flash storage and traditional DRAM on the same module
 - NVDIMM-P (introdotte nel 2018)
 - Persistent Memory (PMEM)
- Storage Class Memory (SCM)
- CXL attached memory
- MMIO
- High Bandwidth Memory (HBM)

Introduction to Persistent Memory

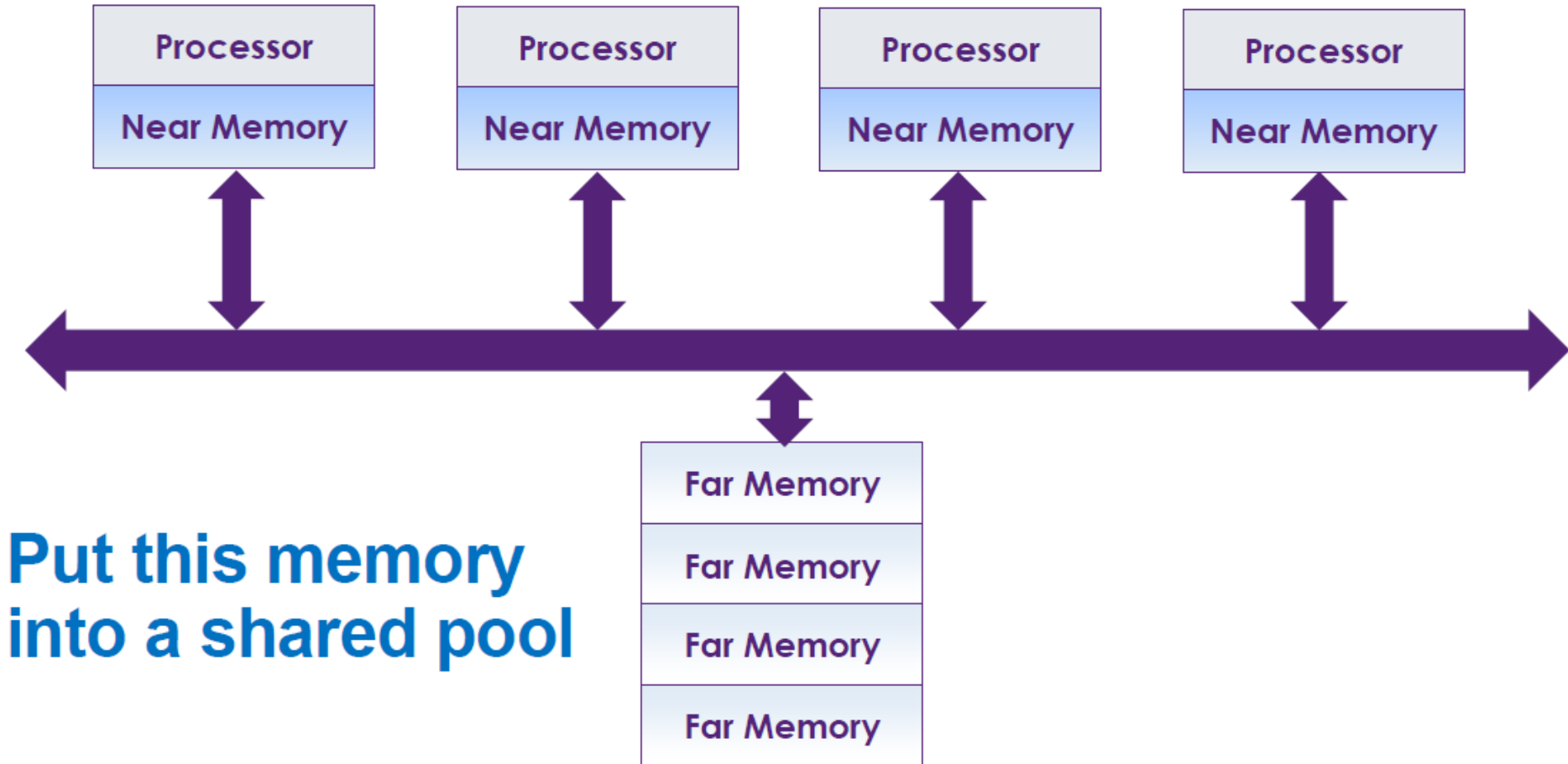
- Memory & Storage is moving closer to the CPU
- Memory & Storage Speeds are increasing
- Persistent Memory (PM) is the newest class of storage
- PM sits on the Memory bus and has:
 - Low Latency
 - High Speed
 - Persistence
 - Cache Coherence



Big Memory



Moving from a von Neumann to a Memory-Centric Compute Model

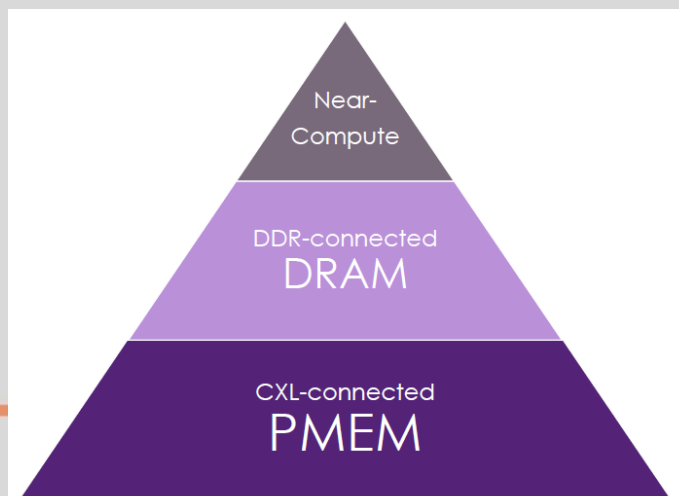


Memory Interconnect Types

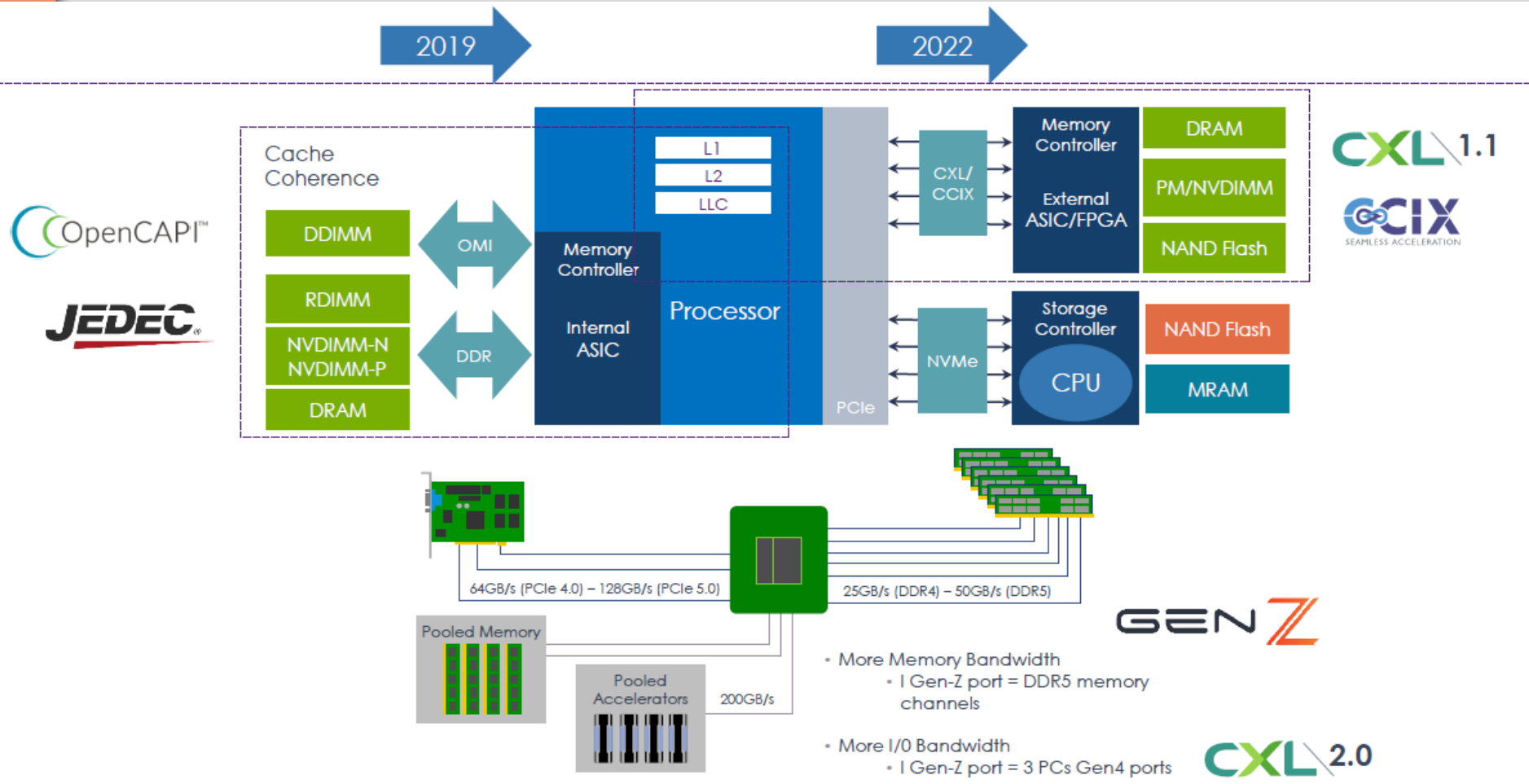
- “Near Memory”
 - DDR good for smaller systems
 - NVDIMM-N e NVDIMM-P
 - HBM fast but restrictive and costly
 - OMI for both high speed and large capacities
- “Far Memory”
 - PCIe / NVMe
 - CXL / CCIX
 - Pools heterogeneous memories
 - Mixed latencies and data rates
 - Gen-Z to connect storage boxes and racks



Evoluzione tecnologica



Memory and Accelerator Alignment with Fabrics

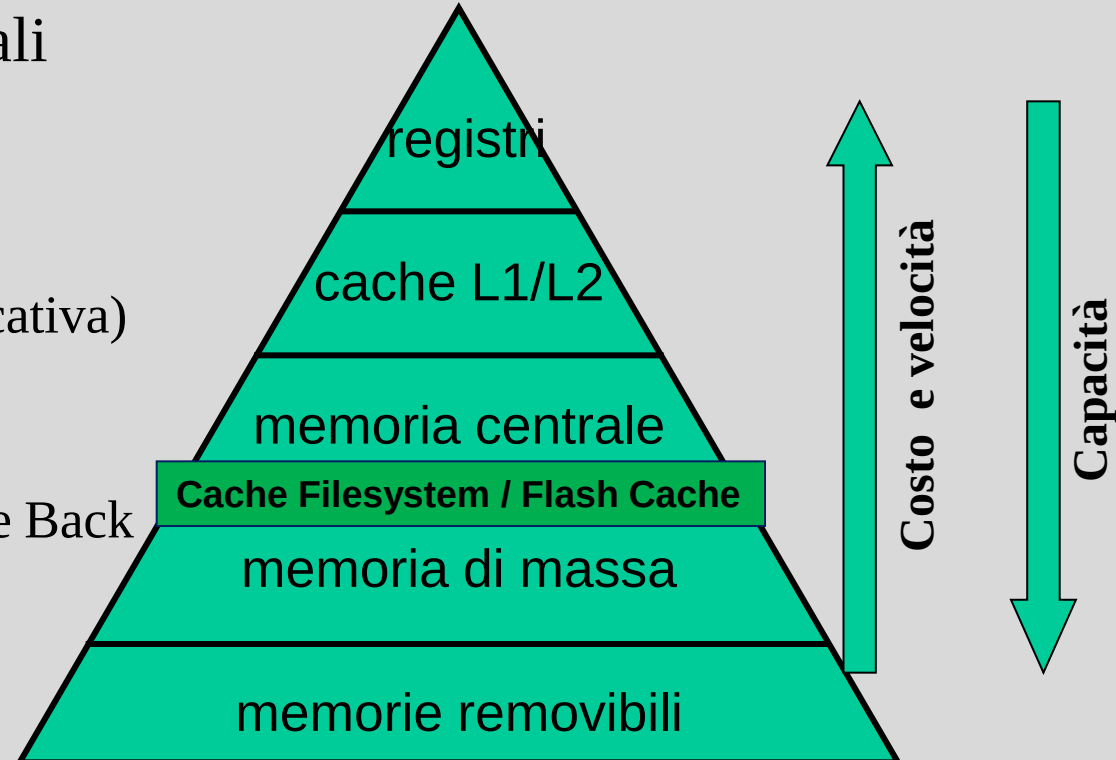


Storage tiers

- Tier 0
 - Storage (primario) con prestazioni paragonabili alla RAM
 - Soluzioni PMEM
- Tier 1
 - Storage primario veloce
 - Prestazioni degli SSD
- Tier 2 o 3
 - Storage primario o secondario
 - Prestazioni buone in lettura
 - Non adatto per applicazioni business critical
- Archive tier
- Hierarchical storage management (HSM)

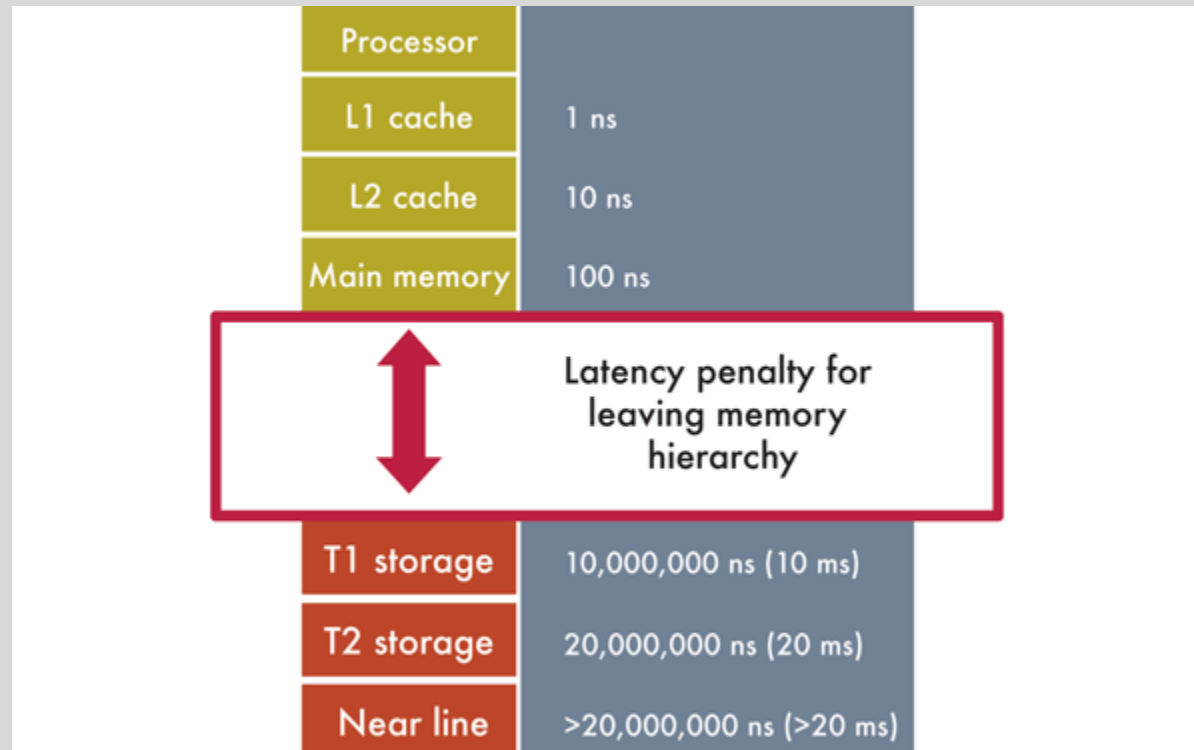
Scelta della cache

- Elemento fondamentale per minimizzare i colli di bottiglia
 - tra CPU e memorie
 - L2 o L3 da 512k e 12MB (condivisa o dedicata per core?)
 - tra disco e memorie
 - ReadyBoost di Vista e “modulo Robson”
- Parametri fondamentali
 - quantità e velocità
- Tipi di cache
 - hardware
 - Software (SO o applicativa)
- Cache policy
 - Read Ahead
 - Write Through / Write Back
- Tempi di accesso
 - RAM: 10 ns
 - HD: 10ms



Disk cache

- Tipicamente implementata dal sistema operativo
- Memorie Flash locali
 - Bus SATA/SAS
 - Bus PCIe
 - Bus memoria
 - NVDIMM/PMEM



Caratteristiche dello storage

- Affidabilità (Reliability), stabilità (Stability), disponibilità (Availability) e ripristinabilità (Recoverability)
 - Variabile a seconda della tecnologia
 - Soluzioni RAID
- Scalabilità (Scalability)
 - Crescita di capacità
 - Crescita della densità
- Prestazioni (Performance)
 - Dischi più veloci
 - Bus più veloci
- Manutenibilità (Manageability) e semplicità
 - Formati standard e hot-plug (nei server)
 - Nuovi formati
- Sicurezza (Security)
 - Dischi SED

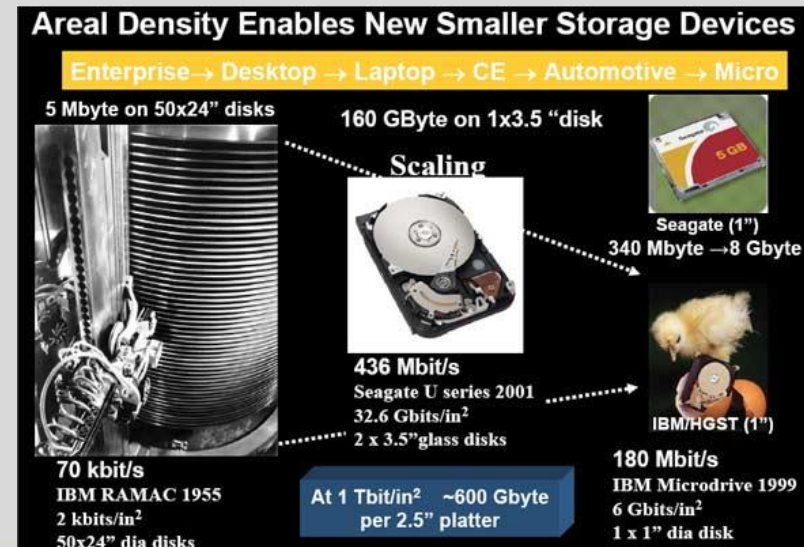
Scelta dello storage secondario

- Tipi di dischi
 - HDD e SSD
- Tipi di storage
 - DAS (Direct Attached Storage)
 - NAS (Network Attached Storage)
 - dischi collegabili in rete Ethernet o Wi-Fi
 - SAN (Storage Area Network)
 - rete ad alta velocità/affidabilità dedicata allo storage
 - Nastri magnetici
 - Altre soluzioni
- Costi e convenienza?



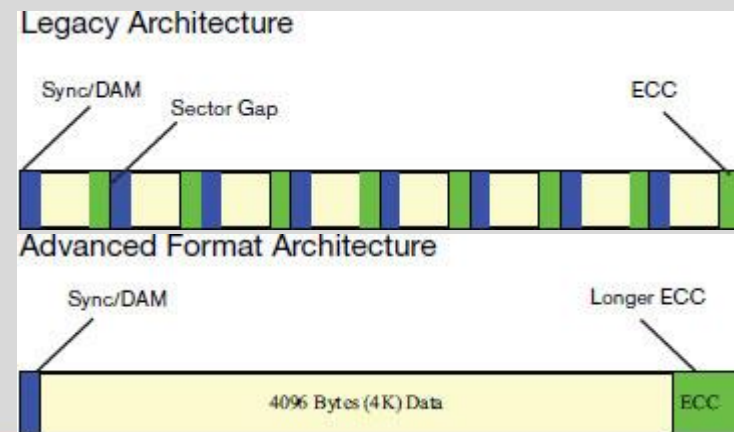
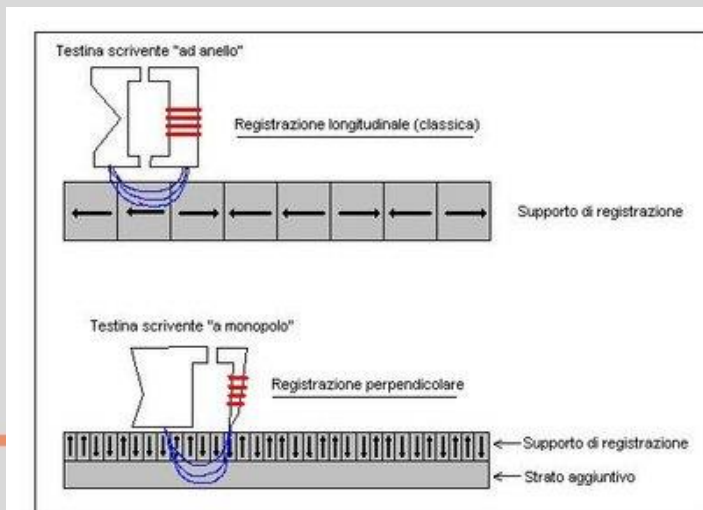
HDD

- Capacità in buona crescita
 - SATA sui 20 TB (2022)
- Pochi produttori
- Prestazioni in crescita limitata (HDD)
- Prestazioni degli HDD determinate da
 - densità di memorizzazione
 - velocità delle testine
 - velocità di rotazione (RPM)
 - numero di canali a disposizione
 - cache integrata/tipo di controller
 - transfer rate
 - tipo di file system
- Affidabilità determinata da
 - tecnologia
 - temperatura di esercizio
 - ...



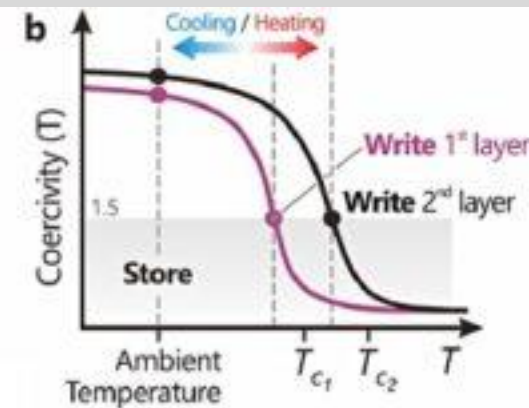
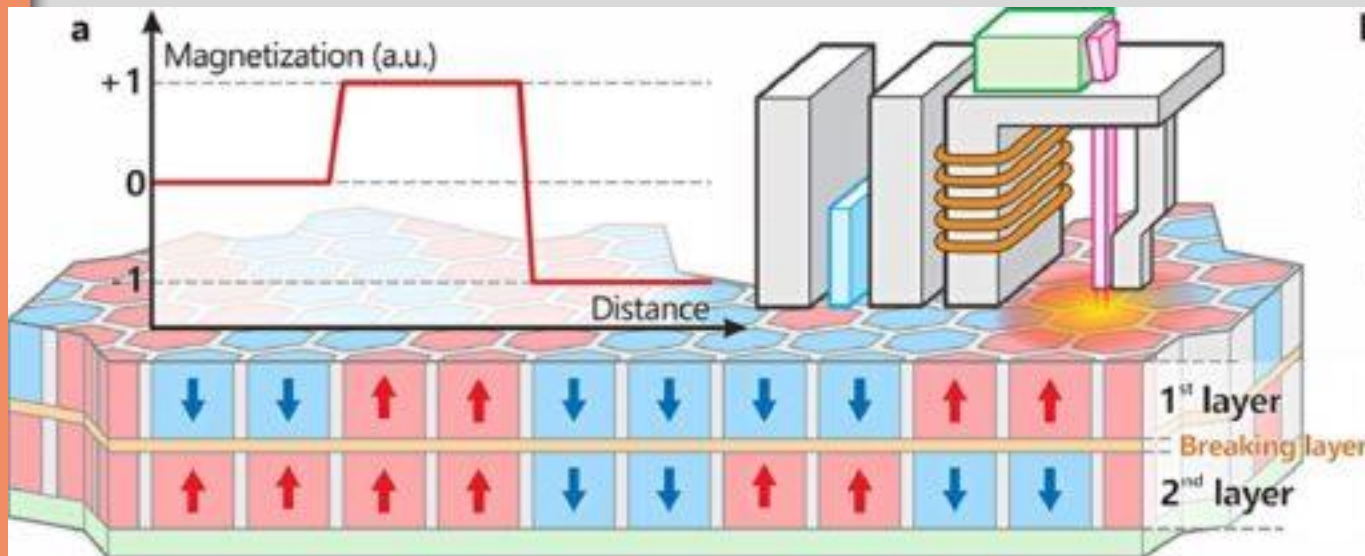
HDD – Nuove tecnologie

- Nuove tecnologie per HDD meccanici
 - Perpendicular Recording (2006)
 - Western Digital Advanced Format (2011)
 - HAMR (Heat-assisted magnetic recording)
 - MACH.2 (doppie testine)
 - Bit Patterned Media e SOMA (Self Ordering Magnetic Array)
 - Blocchi da 4K
 - http://i.dell.com/sites/doccontent/shared-content/data-sheets/Documents/512e_4Kn_Disk_Formats_120413.pdf



HDD – Technologie future

- <https://blocksandfiles.com/2024/04/08/multi-level-hamr/>



Crescita degli HDD

The Future Brings Larger Capacity Steps

Fastest ramping nearline products in PMR while introducing HAMR at multiple capacity points

CY15 CY16 CY17 CY18 CY19 CY20 CY21 CY22 CY23 CY24 CY25 CY26

PMR
MACH·2

16TB

18TB

20TB

HAMR
MACH·2

14 – 20+TB

30+TB

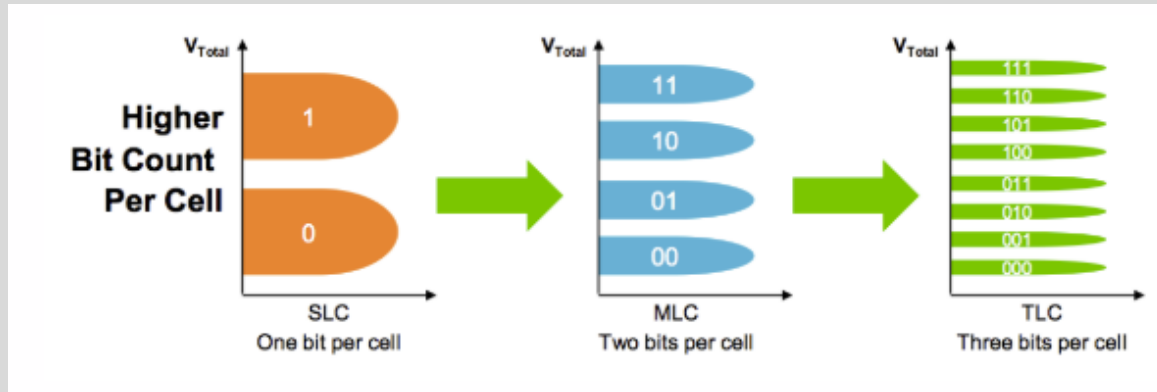
40+TB

50+TB

Source: Seagate Roadmap

SSD

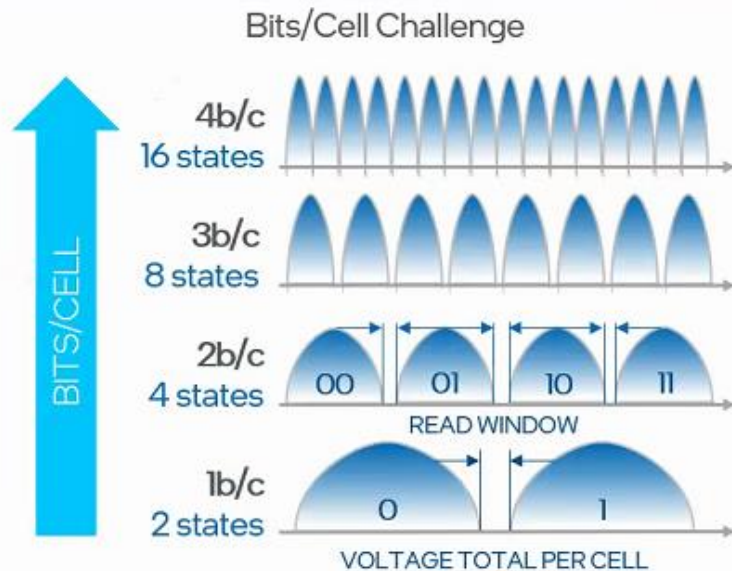
- Alternativa ai dischi HDD
 - Solid State Drive (SSD) e tecnologie Flash
 - Ottima densità dei dati (fino a 100TB)
 - Diverse tecnologie
 - NAND
 - SLC (Single Level Cell)
 - MCL (Multi-Level Cell)
 - TLC (Triple Level Cell)
 - QLC (Quad Level Cell)



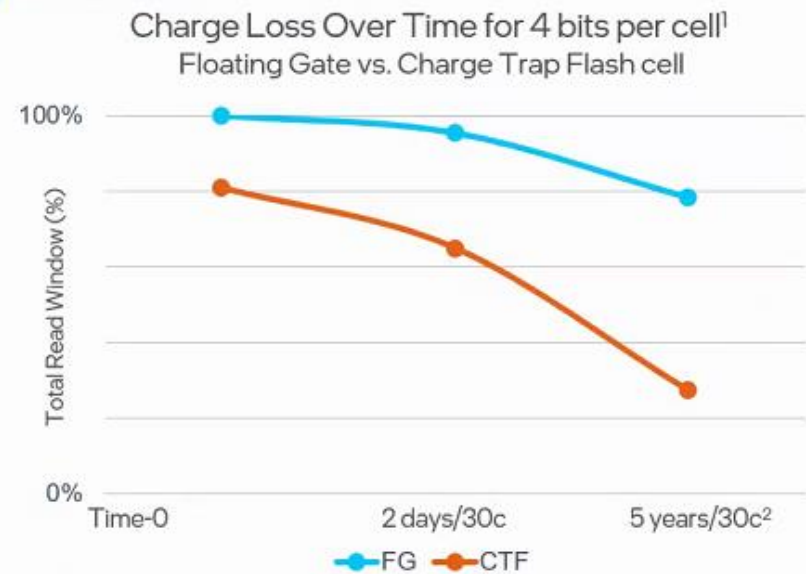
- 3D XPoint
 - Optane

SSD NAND

Floating Gate NAND Delivers High Data Retention



- Smaller read window creates accuracy challenge
- Even smaller amounts of leakage can create read inaccuracies



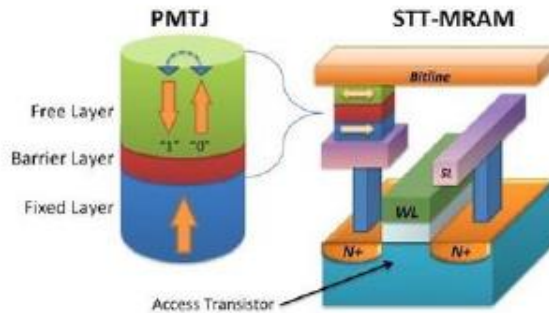
- Floating gate NAND cell superior to charge trap flash for data retention → better QLC capability

¹Source: Measurements performed on components from QLC SSDs using Floating Gate and Charge Trap Flash technology. Measurement platform used was Teradyne Magnum. Memory test systems, and programming using random patterns and margins were quantified using customer commands. Data measured in 08/2019. Results may vary.

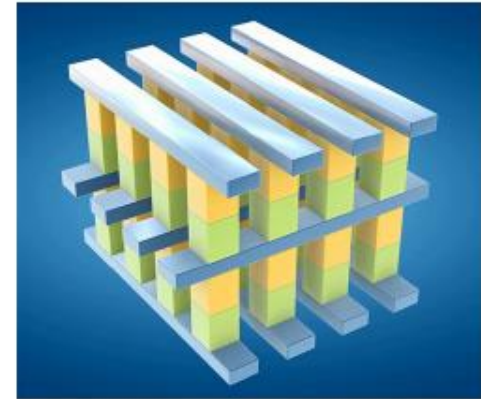
²Ea assumed = 1.0eV

Lots of Emerging Memories

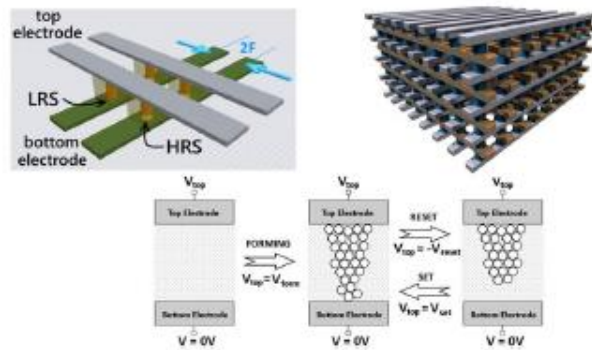
MRAM



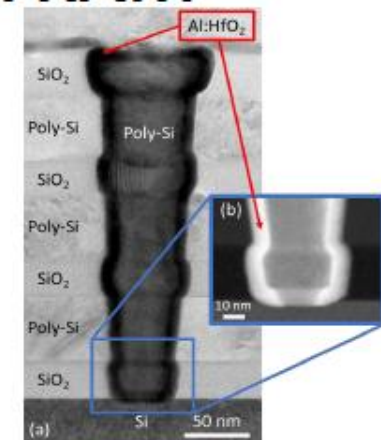
PCM



ReRAM



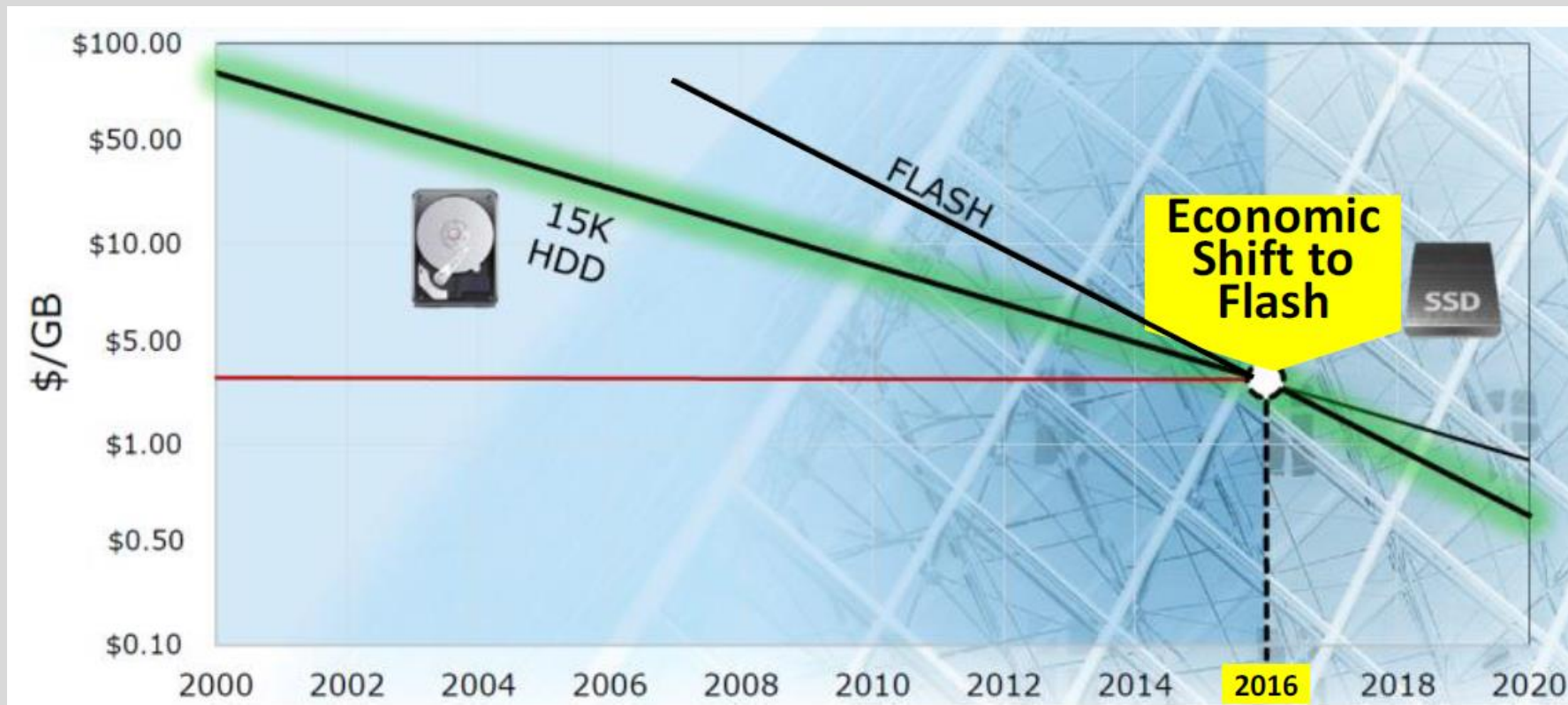
FRAM



Coughlin Associates. All Rights Reserved.

HDD vs. SSD

- Più capacità e prestazioni
- Costo per TB conveniente in molti casi



Disk bus

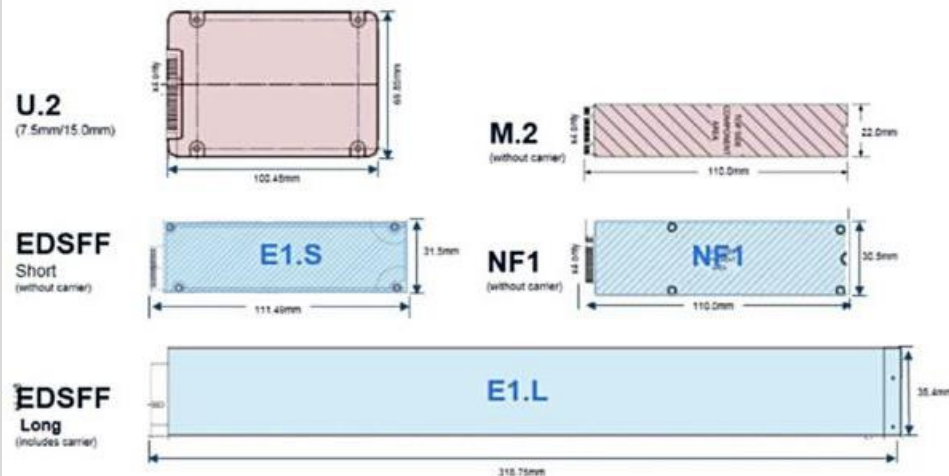
- Diverse tecnologie per il bus
 - P-ATA
 - Max 133 MB/s
 - S-ATA e S-ATA II
 - 150 MB/s (SATA) e 300 MB/s (SATA2)
 - SCSI
 - Ultra-320 (320 MB/s) e dischi a 15K
 - Fibre Channel (FC)
 - 1, 2, 4, 8, 16, 32 Gb/s
 - Serial Attached SCSI (SAS) disk
 - SAS-1: 3Gb/s
 - compatibilità anche con dischi SATA II
 - SAS-2 e SAS-3: 6 e 12Gb/s
 - PCI-e? NVMe? SCM? DIMM?
 - U.2
 - M.2



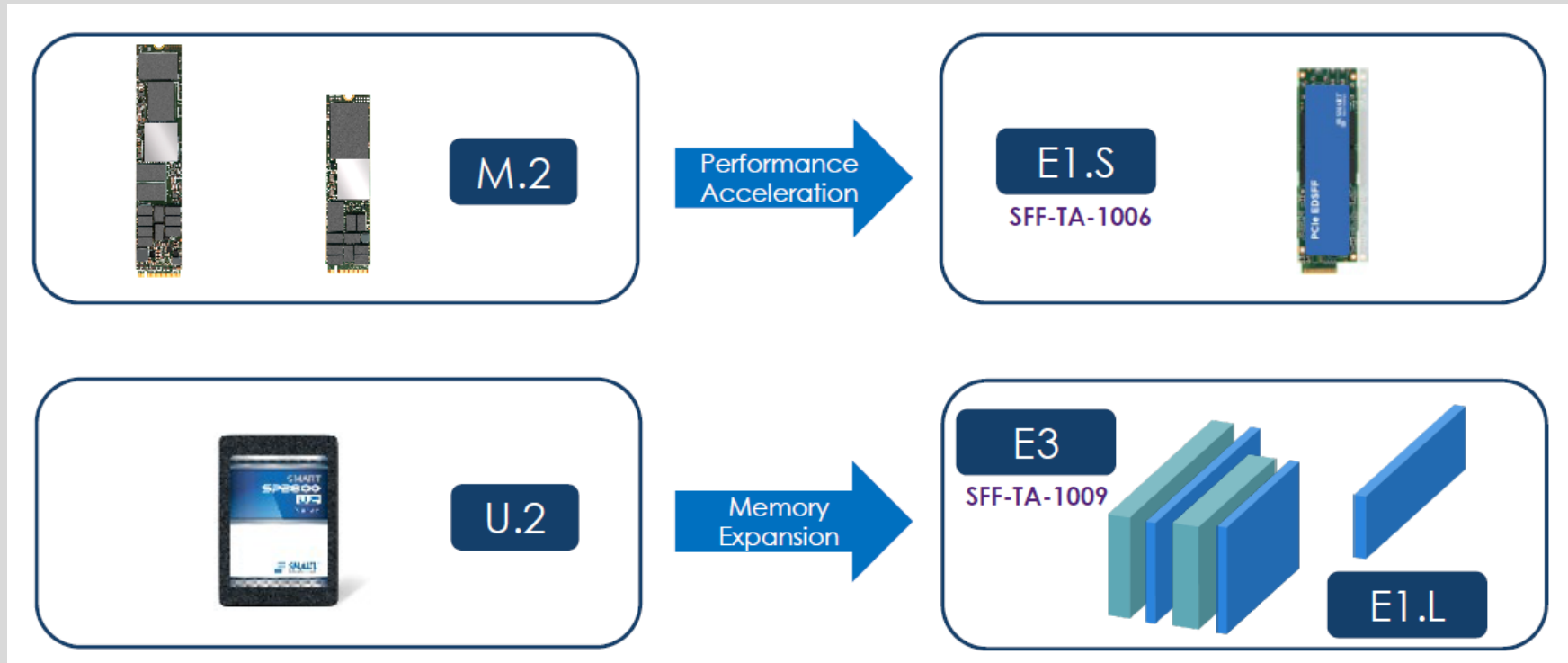
Disk form factor

- HDD case
 - Large form Factor (LFF) 3.5-inch
 - Small Form Factor (SFF) 2.5-inch
- Enterprise & Data Center SSD Form Factor (EDSFF)
 - E1.L (Long)
 - E1.S (Short)
 - Samsung's NF1

NVMe Form Factor Comparison



Flash form factor migration

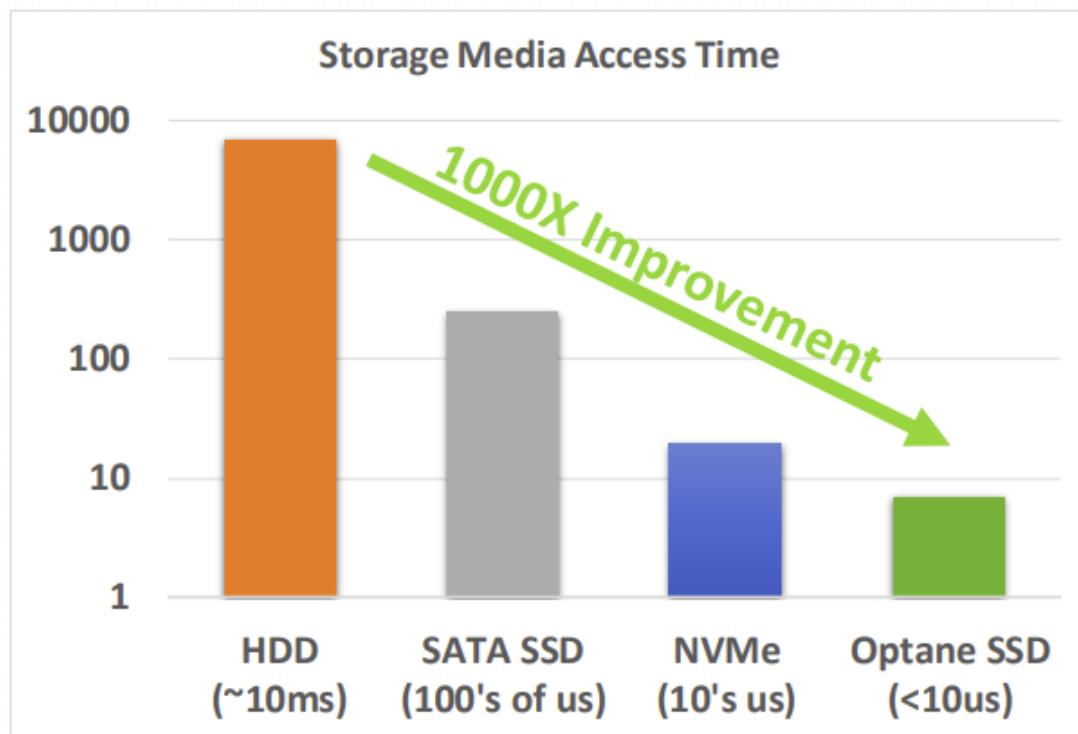


NVM Express

- NVMe defines an optimized queuing interface, command set, and feature set for PCIe SSDs
 - Architected to scale from client to enterprise
- Standardization accelerates industry adoption
 - Standard drivers
 - Consistent feature set
 - Industry ecosystem
 - Development tools
 - Compliance and interoperability testing
- NVMe supporta 64.000 comandi con un massimo di 64.000 queue depth alla volta

Perché NVMe

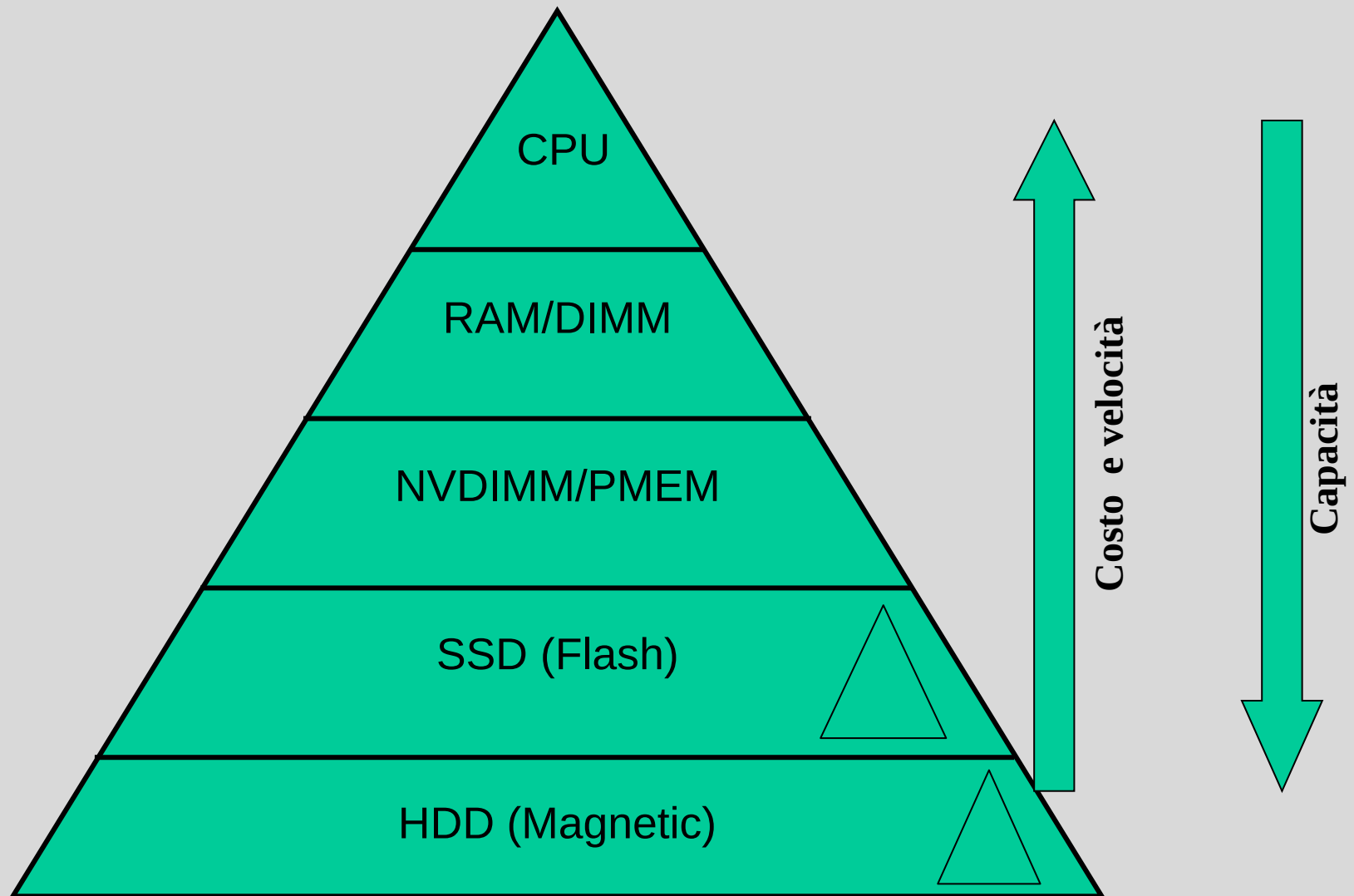
- Nuovo protocollo con Parallelismo migliore e
- Throughput
 - 3 NVMe SSD possono saturare link 100 Gbps
- Latenza migliorata
 - 1,000x vs HDD



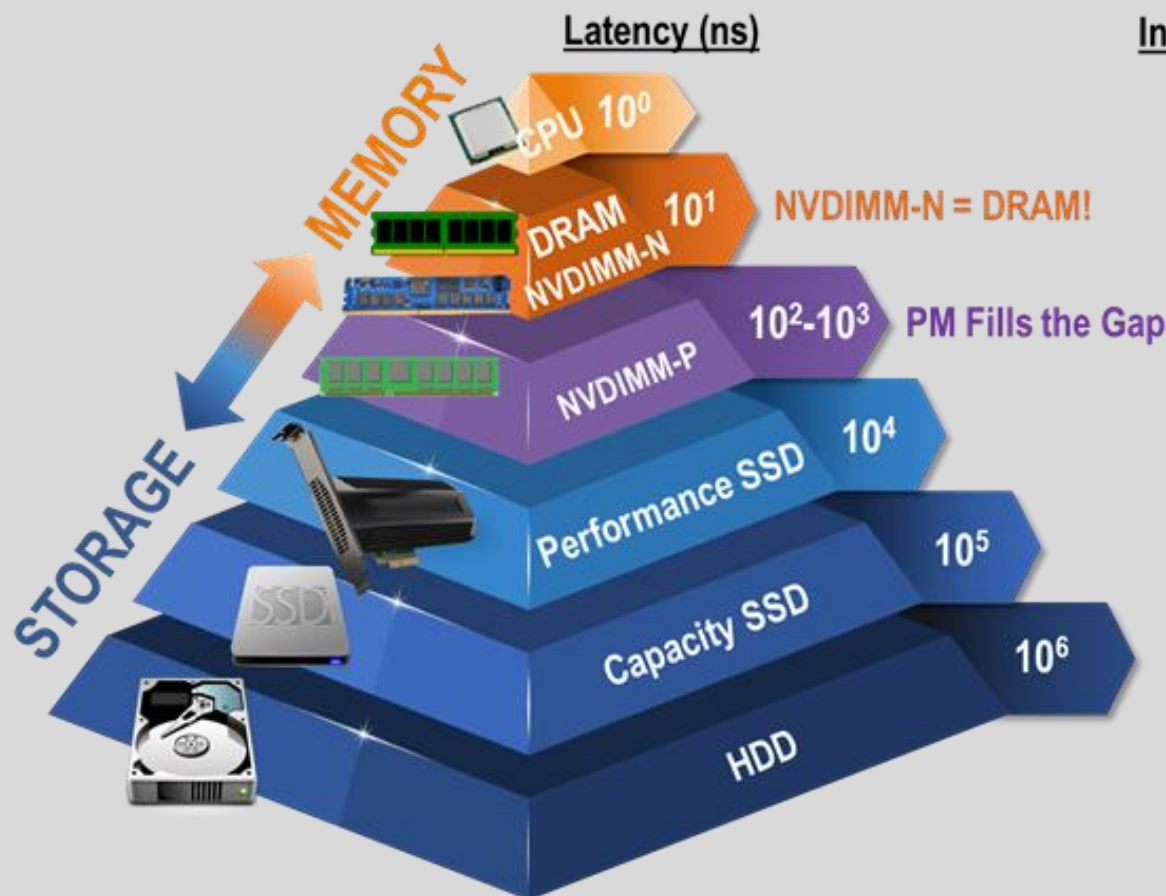
Sempre sui dischi flash

- Problemi aperti che aumentano la latenza
 - Workload isolation
 - Write Amplification Factor (WAF)
 - Background processes
- Soluzioni esistenti
 - ZNS (Zoned Name Space)
 - Risolve WAF e isolation
 - «Use flash like a bunch of tape»
 - FDP (Flexible Data Placement)
 - Software defined flash

Piramide delle memorie



Piramide delle memorie



In Human Observable Terms

Seconds



Minutes



Hours



Days

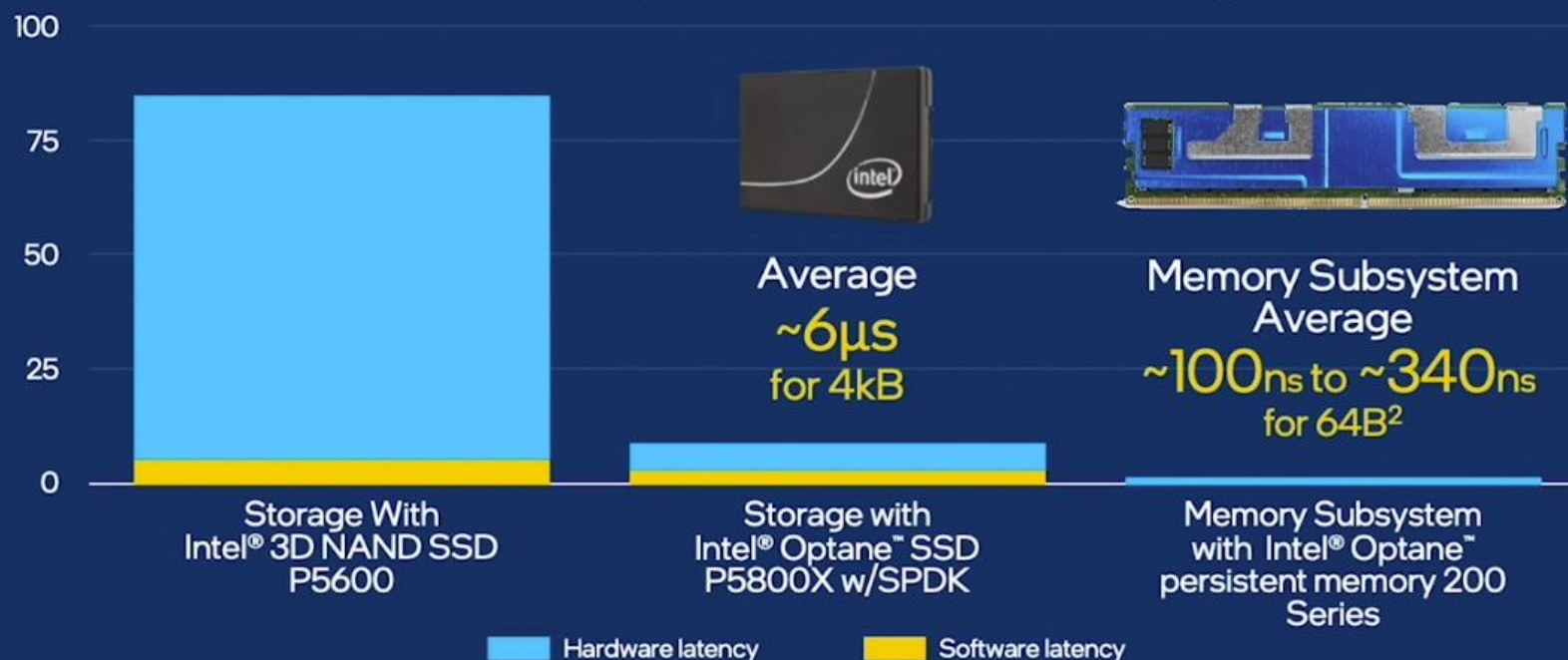


Weeks



More to be Gained by Being on Memory Bus

Average Lightly Loaded Random Read Latency¹

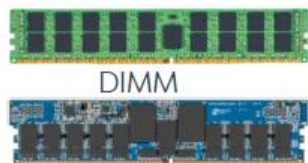


Performance results are based on testing as of September 25, 2020. Performance varies by use, configuration and other factors, for details, see [Slide 14](#). Learn more at www.intel.com/PerformanceIndex.

Performance

SNIA PERSISTENT MEMORY
+ SUMMIT 2021
COMPUTATIONAL STORAGE

Direct attached
(Parallel Bus)
100's of GB



DIMM
NVDIMM-N, NVDIMM-P
(Persistent)

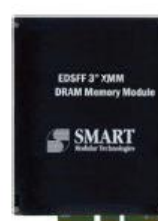
Serial attached
and PCIe attached
100's of GB to TB's



E1.S (x4)



E1.S (x8)



E3.S (x16)



PCIe AIC (x16)



OpenCAPI
DDIMM

Network Attached
TB's to PB's



ZMM

	DDR DIMM	E1.S 1C (x4)	E1.S 2C (x8)	E3.S AIC (x16)	E3 with x8 (2C)	Across network
Current Generation *	DDR4@3200 25.6GB/s	PCIe-Gen4-x4 7.8GB/s	PCIe-Gen4-x8 15.7GB/s	PCIe-Gen4-x16 31.5GB/s	OMI 25.6 GB/s 8 lanes	RDMA (Fabric and work load dependent)
Future Generation **	DDR5@4800 63.0GB/s	PCIe-Gen5-x4 15.7GB/s	PCIe-Gen5-x8 31.5GB/s	PCIe-Gen5-x16 63.0GB/s	DDR5 DDIMM (TBD) GB/s	Gen-Z, NVMe-oF (TBD)

* Source(s): https://en.wikipedia.org/wiki/PCI_Express#History_and_revisions
https://en.wikipedia.org/wiki/DDR4_SDRAM#JEDEC_standard_DDR4_module
<https://www.smartm.com/media/press-releases/smart-modular-to-showcase-its-ddr4-differential-dimm-at-the-flash-memory-summit>

** Source(s): <https://www.tomshardware.com/news/ddr5-6400-ram-benchmarks-major-performance-gains-ddr4>

IOPS

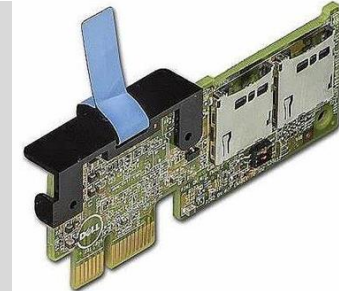
- IOPS (Input/Output Operations Per Second)
 - Misurazione comunemente usata per definire le prestazioni di una memoria secondaria
 - Ma anche per dimensionare lo storage
- Ogni tipo di disco ha un limite di IOPS
 - Random Read or Write IOPS
 - Average number of random read or write I/O operations per second
 - Sequential Read or Write IOPS
 - Average number of sequential read or write I/O operations per second
 - Total IOPS
 - Total number of I/O operations per second (when performing a mix of read and write tests)
- Come ottenere IOPS maggiori?

RAID

- Disco «logico»
 - Basato su diversi dischi fisici
 - Con più ridondanza e/o prestazioni
- Implementazione del RAID
 - Hardware
 - Software
- Alternative al RAID?

Boot device

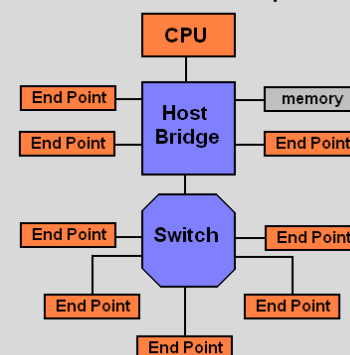
- Diverse soluzioni basate su dischi
 - Dischi locali (su controller RAID)
 - Schede ad hoc
 - BOSS card
 - ISDM
 - ...
 - Chiavi USB e SATADOM
- Boot from SAN
- Boot from LAN
 - PXE Boot



Scelta del bus di sistema

- PCI Express (PCI-e) (Peripheral Component Interconnect Express)
 - Bus seriale che sostituisce PCI, PCI-X e AGP
 - PCI Express 2.0 (2007)
 - Fino a 500 MB/s per ogni linea dati
 - PCI Express 3.1 (2010)
 - Fino a 1 GB/s per ogni linea dati
 - PCI Express 4.0 (2017)
 - Quasi 2 GB/s per ogni linea dati
 - Fino a 16 canali per scheda | 31,5 GB/s totali
 - PCI Express 5.0 (2019)
 - Banda di 63 GB/s con un collegamento 16x
 - PCI Express 6.0 (2022)
 - Banda di 121 GB/s con un collegamento 16x
- Problemi di compatibilità?
- Backplane Ethernet (IEEE P802.3ap)
- Altri bus
 - CXL, NVMe, NVDIMM, ...

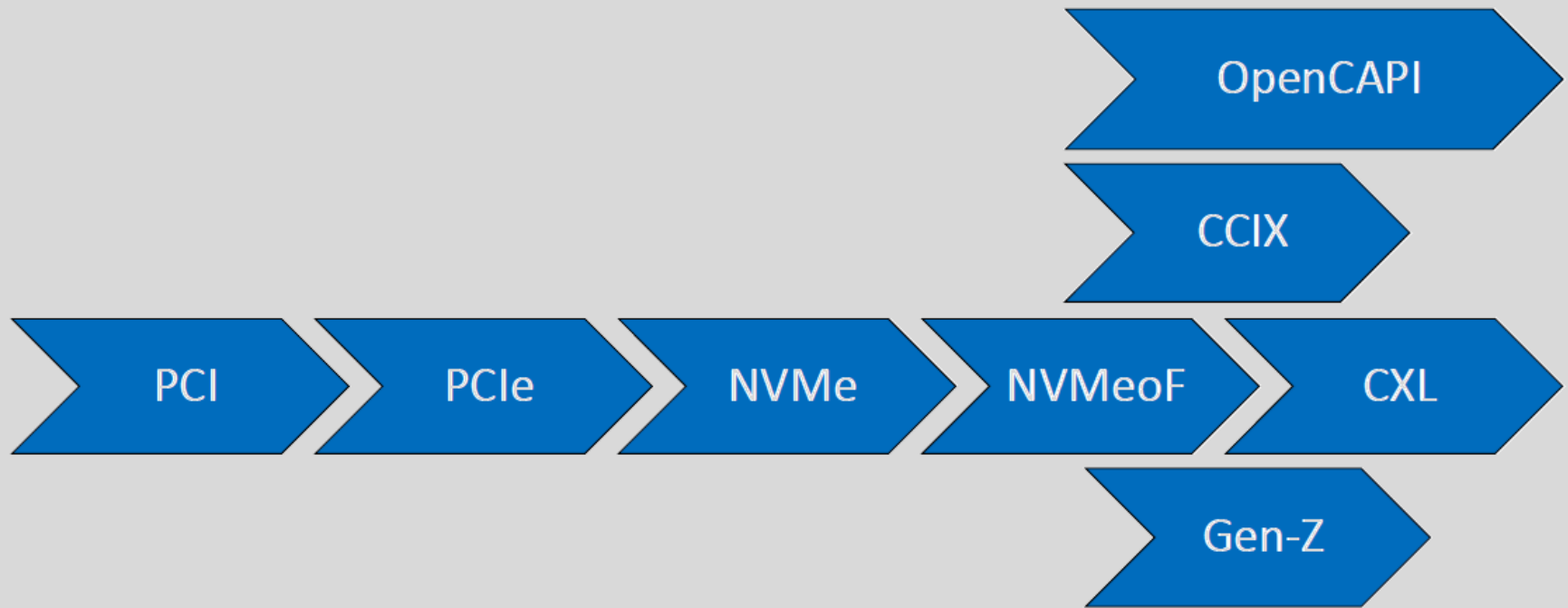
Struttura Bus PCI-Express



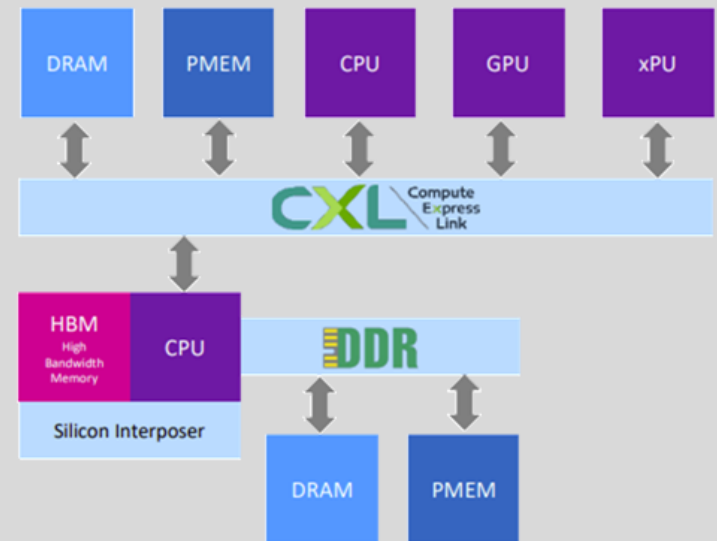
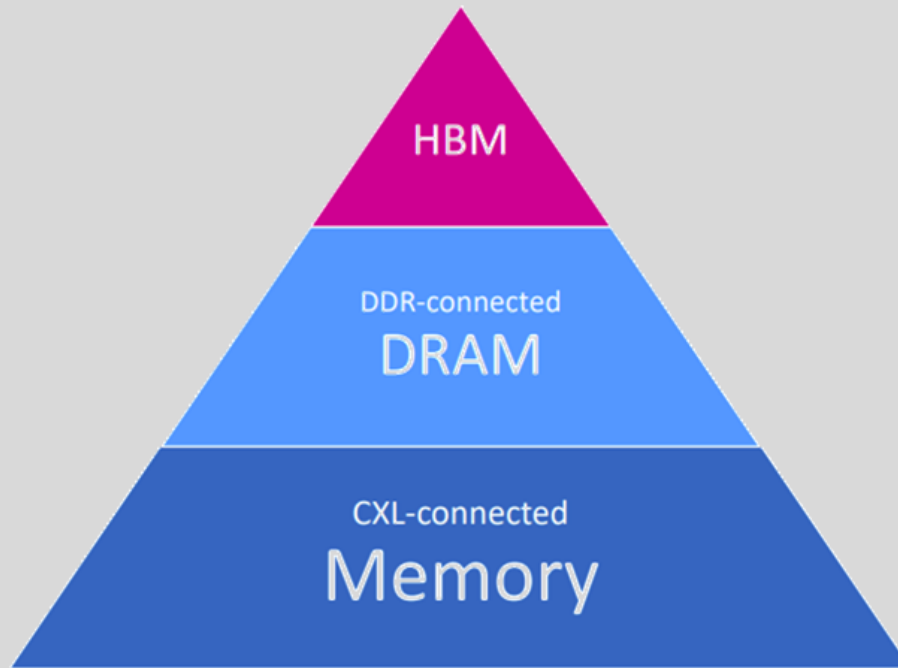
PCI EXPRESS



Evoluzione dei bus di sistema



CXL is a game changer



But hardware Innovation Alone
Is Not Enough

Scelta del bus esterni

Thunder bolt 3	USB4	USB 3.0 3.1 3.2	USB 2.0	eSATA	Giga Ethernet
Hot-plug	Hot-plug	Hot-plug	Hot-plug	Hot-plug	
	LowPower	LowPower	Alimentato	Power eSATA	PoE
40 Gb/s <i>teorici</i>	40 Gb/s <i>teorici</i>	4,8 / 10 / 20 Gb/s <i>teorici</i>	480 Mb/s <i>teorici</i>	3 Gb/s	10 Gb/s 40 Gb/s

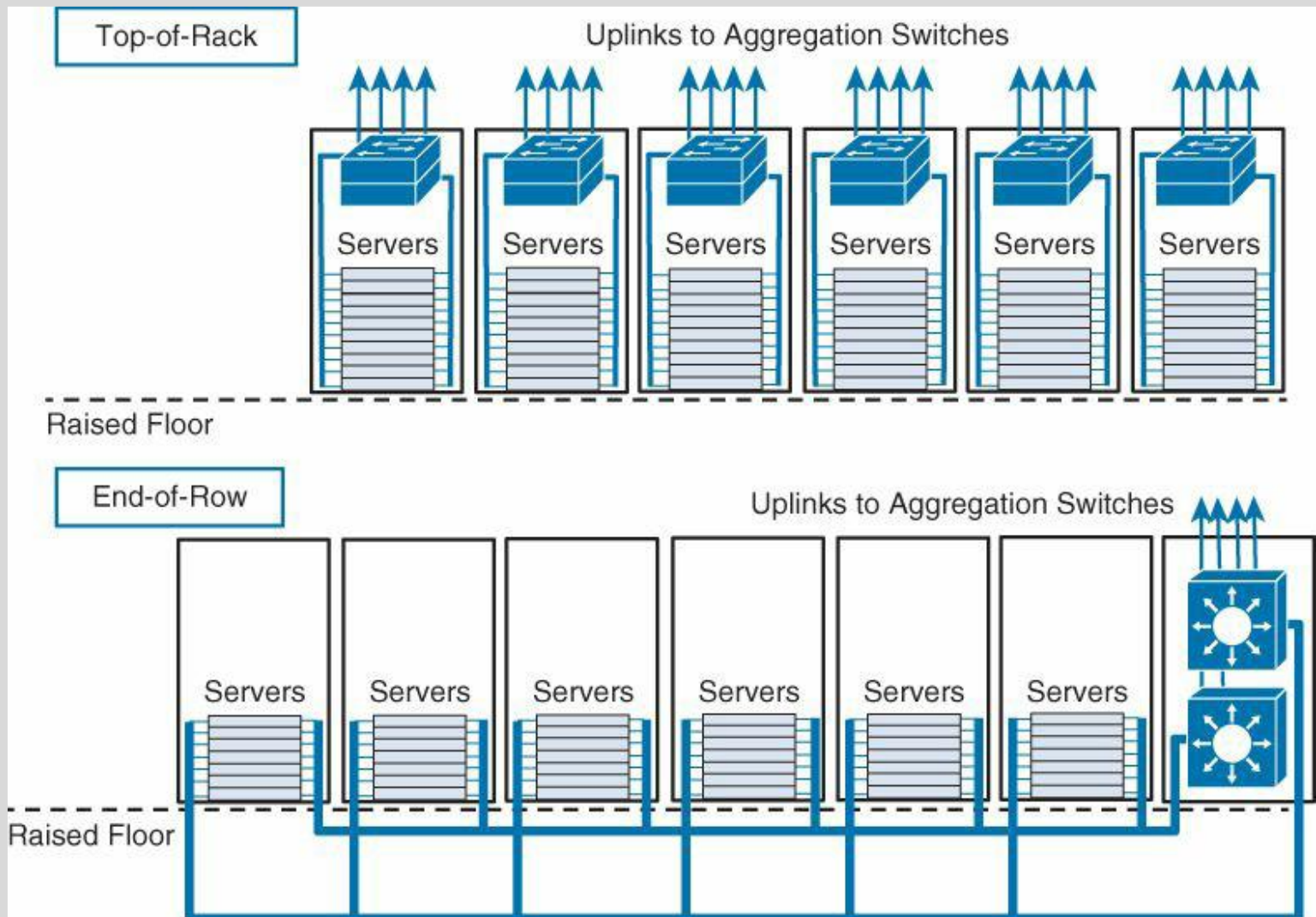
Altri “bus” esterni: seriale, parallela, SCSI, SAS, FC, ...



Scelta delle schede di rete

- Famiglia di standard IEEE 802.3
 - 1 Gpbs
 - Fibra - IEEE 802.3z
 - Rame - IEEE 802.3ab
 - 10 Gbps
 - Fibra - IEEE 802.3ae
 - Rame - IEEE 802.3an
 - 40 Gb/s e 100 Gb/s
 - IEEE 802.3ba
 - 25 Gbps/s
 - IEEE 802.3bq e 802.3by
- Multi standard?

Server connection models



LOM versus NDC

- LAN-on-Motherboard (LOM)
- Network Daughter Card (NDC)
- Add-on NIC

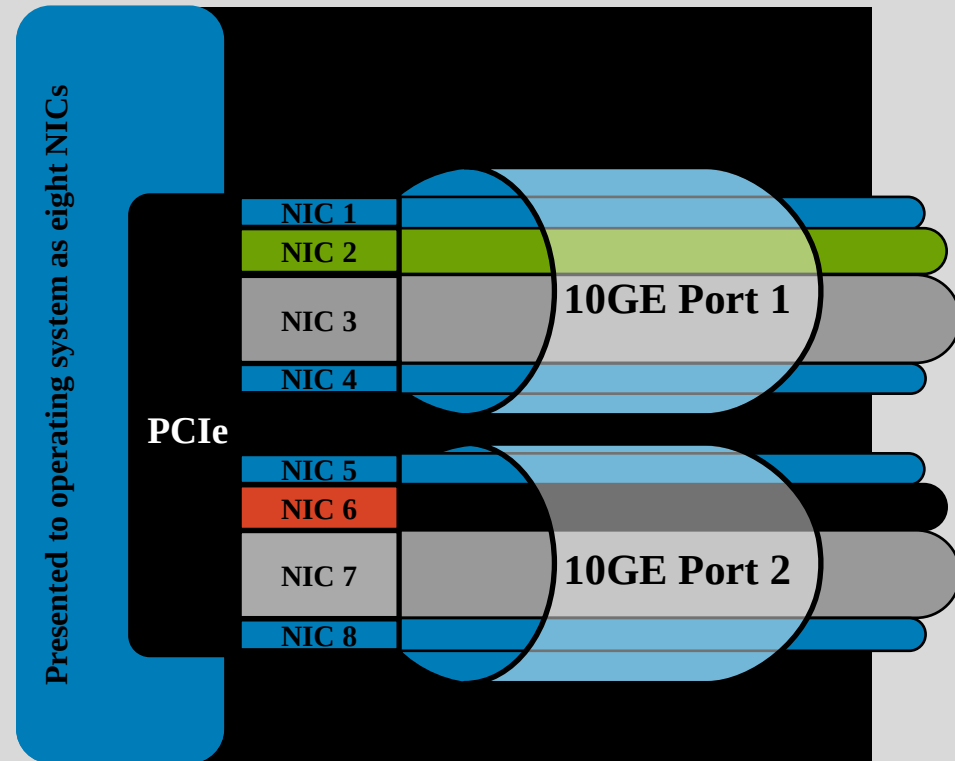


Select Network Adapter



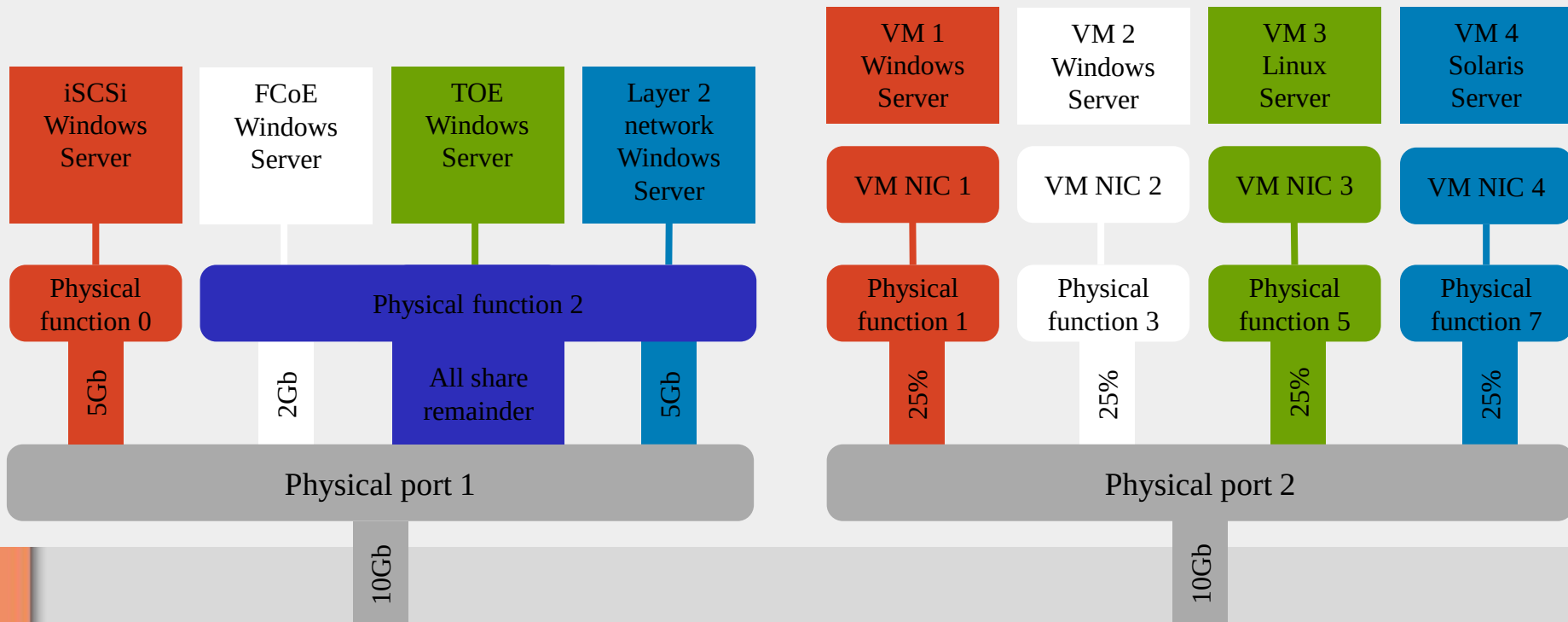
Partizionamento delle schede

- Physical functions
 - Logical functions
 - Virtual functions
 - Offload
-
- Tecniche
 - NPAR
 - SRV-IO



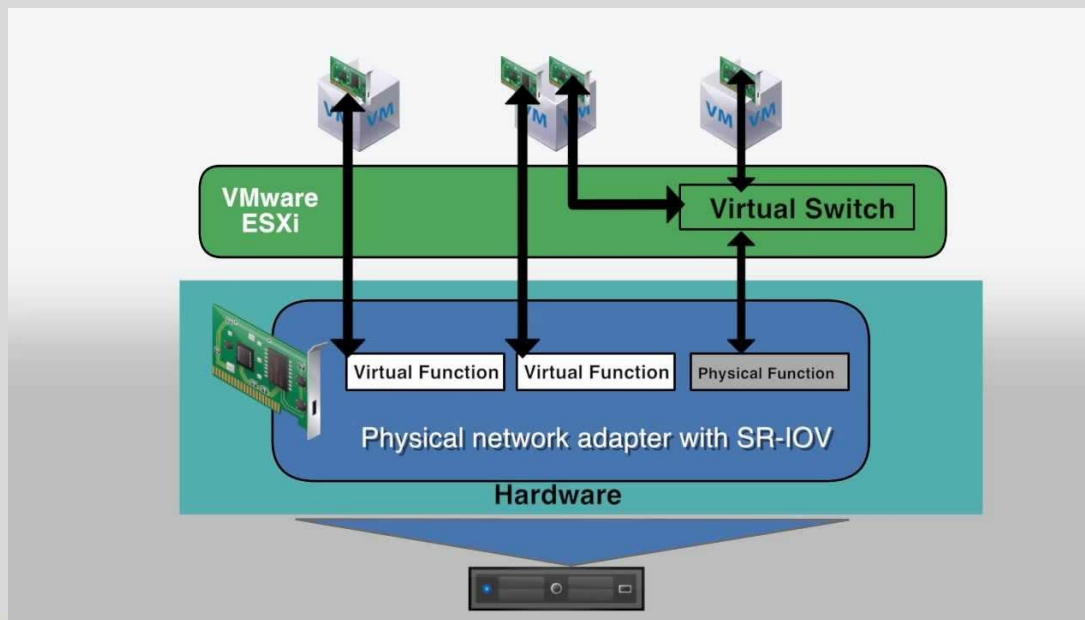
NPAR

- Network Partition
 - create multiple native Ethernet interfaces that share a single physical port



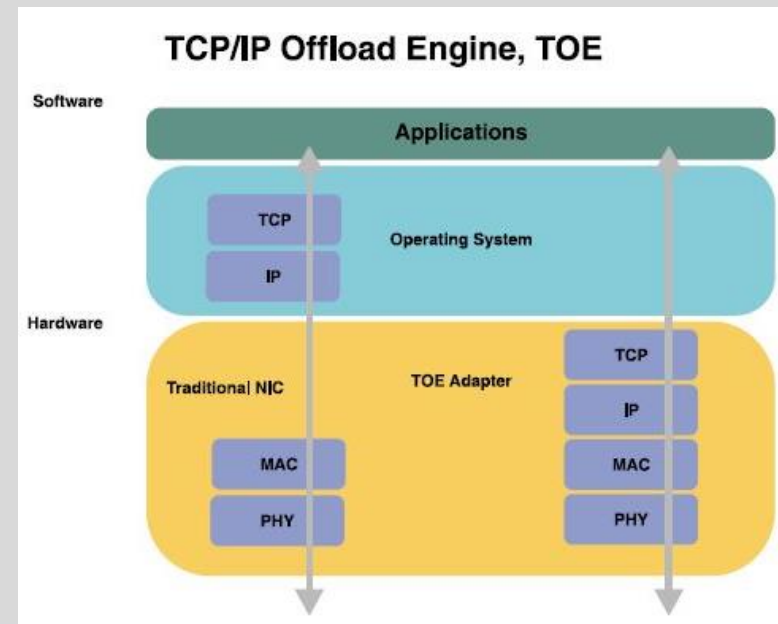
SR-IOV

- Single Root – IO Virtualization
 - standard that can present single PCIe device (this is called Physical Function) as multiple independent PCIe devices (each one is called Virtual Function) to Operating Systems and hypervisors



“Accelerazione” hardware

- Ottimizzazioni
 - Jumbo Frame
- Hardware offload
 - TCP Offload Engine (TOE)
 - Intel I/O Acceleration Technology (I/OAT)



SmartNIC

- FPGA on NIC
- Software-Defined, Hardware-Accelerated Data Center Infrastructure on a Chip
- NVIDIA® BlueField® data processing unit (DPU)



■ ASIC Based

- Excellent price-performance
- Vendor development cost high
- Programmable and extensible
 - Easy to program but flexibility is limited to pre-defined capabilities



■ FPGA Based

- Good performance but expensive
- Very difficult to program
- Workload specific optimization



■ SOC Based

System on Chip - NIC + CPU

- Good price-performance
- C Programmable Processors
- Highest Flexibility
- Easiest programmability

Evoluzione delle NIC

Capability	Workloads Accelerated	Foundational NIC	Intelligent NIC (iNIC)	SmartNIC
Entry Level Virtualization and Data Movement				
TCP/IP Acceleration	Enterprise workloads	✓	✓	✓
NIC Virtualization (SR-IOV)	Enterprise workloads	✓	✓	✓
Network Virtualization (VXLAN)	Multi-tenant workloads	✓	✓	✓
Data Transport Acceleration & Programmable Data Plane				
RoCE Acceleration	VM, Storage, Big Data, AI/ML		✓	✓
DPDK	Virtualized Network Functions		✓	✓
Spark Acceleration	Big Data		✓	✓
NVMe-over-Fabrics Storage	Storage		✓	✓
OVS Hardware Acceleration	Efficient, Scalable Virtualized Apps		✓	✓
QoS and ACL Acceleration	Web Servers / Content Distribution Networks		✓	✓
Flow monitoring/reporting	Visibility, Network Packet Broker, IBN		✓	✓
Flow match/action engine	Software Defined Networking		✓	✓
Fully Programmable Data Plane	Network Function Virtualization		✓	✓
ASAP ² Virtual Switching/Routing	NAT, Load Balancing, stateless firewall		✓	✓
Smart Networking & Virtualization				
Virtual Switch Policy Engine	OVS Control Plane			✓
Analytics engine	DPI, Network Monitoring and Diagnostics			✓
Container Acceleration	Various (AI/ML, Big Data, Analytics)			✓
Smart Cloud Virtualization	Server Disaggregation & Resource Sharing			✓
Security, Compression, Network Function Virtualization, Storage				
Public Key Crypto, RNG	Authentication, Key Exchange			✓
Fault Domain Isolation/HA	Bare Metal Cloud			✓
Stateful IP/ACL filtering	Load Balancing IPD/IDS/UTM			✓
Storage: Hashing, ECC, Compression	Erasure Coding, Thin Provisioning, Dedup			✓
Encryption/Decryption	Data at Rest or on-the-fly			✓
Security VNF Offloading	Firewall, IDS, IPS, Anti-malware, Anti-DDoS			✓